UNIT I

Introduction to the Development of Industrial Safety & Management

history & development | Factories Act | safety & productivity | safety organizations | committees | role of management & Govt.

I. Introduction - basic terms

Industrial safety is the branch of engineering/management that identifies the hazards in an industry, evaluates the risk and puts controls in place so that **workers, machines, materials and the environment** are protected from harm. A safe plant is also an efficient plant.

Term	Meaning (easy definition)
Hazard	Any source/situation with the potential to cause injury, illness, damage or loss (e.g. oil on floor, exposed wire).
Risk	Chance (probability) that a hazard actually causes harm x how severe the harm is. Risk = Probability x Severity
Accident	An unplanned, uncontrolled event that results in injury, ill-health or damage.
Incident	Any unwanted event that could / did cause harm; wider term than accident.
Near miss	Unplanned event that did NOT cause injury or damage but had the potential to - a free warning!
Unsafe act	Human behaviour that deviates from safe practice (no PPE, removing a guard, horseplay, shortcuts).
Unsafe condition	Physical defect in the workplace (no guard, poor lighting, wet floor, faulty tool).
Disabling injury	Injury that prevents the worker from doing normal work for one full day (shift) or more.

Objectives of industrial safety

> To **prevent injuries**, deaths and occupational diseases of workers.

> To protect plant, machinery, raw material and finished goods from damage / fire / explosion.

> To reduce the **cost of accidents** (medical, compensation, lost production, insurance premium).

> To comply with legal duties (Factories Act, rules) and avoid penalties and prosecution.

> To improve morale, goodwill and the public image of the company.

> To build a **safety culture** where every person feels responsible - 'safety is everybody's business'.

II. History & development of industrial safety

2.1 Early period

> **Ancient times**: Code of Hammurabi (c. 1750 BC) fixed penalties for builders whose faulty work killed people; Egyptians & Romans used simple protective measures in mines.

> **Before 1750**: work was done in homes / small workshops (cottage industry); hazards were limited and mostly individual.

> **Industrial Revolution (1760-1840, England)**: steam power, textile mills, coal mines. Unguarded belts & shafts, long hours (12-16 hrs), **child labour**, poor ventilation, no compensation - accidents and diseases rose sharply.

> Public protest and reports led to the first **factory laws**, and employers slowly realised that accidents cost money.

2.2 Milestones in the world

Year	Event / development
1802	UK **Health and Morals of Apprentices Act** - first law on working conditions in cotton mills (cleanliness, ventilation, limited hours).
1833	UK Factory Act - appointed the first **factory inspectors**; banned work by children under 9 in textile mills.
1844-1878	Further Factory Acts - machinery fencing, 10-hour day for women & young persons, consolidated in the Factory & Workshop Act 1878.
1893	Railway Safety Appliance Act (USA) - automatic couplers & air brakes on trains.
1911	Triangle Shirtwaist fire, New York (146 workers died) - locked exits; triggered fire-safety & factory laws.
1913	**National Safety Council** (USA) founded - first voluntary safety body.
1919	**ILO** (International Labour Organization) formed - international labour standards, incl. safety.
1931	H.W. **Heinrich** publishes 'Industrial Accident Prevention' - domino theory, ratio study.
1970	**OSHA Act** (USA) - employers must give a workplace free from recognised hazards. NIOSH created.

Year	Event / development
1974	UK **Health and Safety at Work Act (HSWA)** - goal-setting law, general duty of care, Safety Representatives.
1980s-90s	Bhopal (1984), Chernobyl (1986), Piper Alpha (1988) -> process safety, HAZOP, safety cases, SEVESO rules.
2018	**ISO 45001** - international standard for Occupational Health & Safety Management Systems (replaced OHSAS 18001).

2.3 Major industrial disasters - lessons learned

Disaster	Cause / lesson
Flixborough, UK (1974)	Temporary pipe bypass failed -> vapour cloud explosion, 28 dead. Lesson: **management of change**.
Seveso, Italy (1976)	Runaway reaction released dioxin. Led to the EU SEVESO Directive on major-accident hazards.
Bhopal, India (Dec 1984)	Methyl isocyanate (MIC) leak from Union Carbide plant; thousands died. Lesson: safety systems must work, emergency planning, community right-to-know. Led to **Factories Act amendment 1987** (Ch. IVA hazardous processes).
Piper Alpha, North Sea (1988)	Gas explosion on offshore platform, 167 dead. Lesson: permit-to-work and safety management.
Chasnala mine, India (1975)	Water inrush into coal mine, 375 miners died. Lesson: mine safety, flood protection.
Vizag gas leak, India (2020)	Styrene vapour leak from a polymer plant after lock-down restart; 12 died. Lesson: safe restart & storage of monomers.

2.4 Development of industrial safety legislation in India

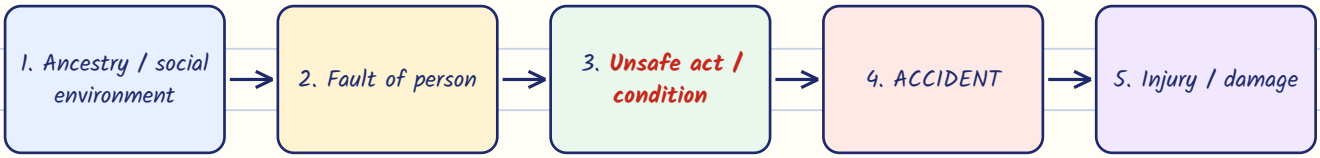
Year	Development
1881	First Indian **Factories Act** - child labour (7-12 yrs) restricted; machinery fencing.
1891, 1911, 1922	Revised Factories Acts - hours of work, women's hours, wider definition of factory.
1923	Workmen's Compensation Act and Indian Boilers Act.
1929-31	**Royal Commission on Labour** (Whitley Commission) - recommended safety measures and inspectors.

Year	Development
1934	Factories Act - 54-hr week, strong inspection system.
1948	**Factories Act, 1948** (in force 1 April 1949) - replaced the 1934 Act; also the ESI Act 1948.
1976	Factories Act amended - extended to wider categories, workers' participation.
1987	Post-Bhopal amendment - Chapter IVA: hazardous processes, site appraisal committee, safety officers, disclosure of information, workers' right to warn.
2020 / 2025	Parliament passed the **OSH Code 2020**; brought into force on 21 Nov 2025.

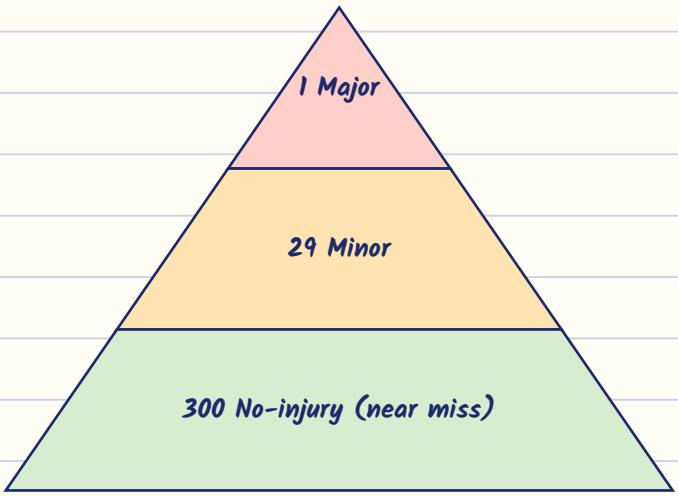
III. Theories of accident causation (basis of safety)

3.1 Heinrich's Domino theory (1931)

Accident is a chain of five dominoes. If any one domino is removed the chain stops. The **key domino** to remove is the **3rd - unsafe act / unsafe condition**.



3.2 Heinrich's accident ratio and Bird's ratio



Heinrich 1 : 29 : 300

1 major injury

29 minor injuries

300 no-injury events

Bird 1 : 10 : 30 : 600

serious : minor : damage : near-miss

88% unsafe acts, 10% unsafe conditions, 2% unavoidable

> Message: **reduce the base** (unsafe acts & near-misses) and the top of the pyramid - the serious injuries - will fall by itself.

> Other theories: **Multiple-causation theory**, **Accident-proneness theory**, **Energy-release theory**

(Haddon), **Human-factor / Ferrell theory**, **Systems & Fault-tree approach**.

IV. Implementation of the Factories Act, 1948

The **Factories Act, 1948** is the main central law in India that regulates **safety, health, welfare, working hours and employment of young persons** in factories. It is administered by the **State Governments** through the **Chief Inspector of Factories** and the **Directorate General Factory Advice Service & Labour Institutes (DGFASLI)** at Centre.

4.1 Important definitions (Sec. 2)

Term	Meaning under the Act
Factory [2(m)]	Premises where **10 or more** workers work using power, or **20 or more** workers working without power, on any day of the preceding 12 months, in a manufacturing process.
Worker [2(l)]	Person employed (directly / through agency, with or without wages) in a manufacturing process or related work; excludes members of armed forces.
Occupier [2(n)]	Person who has ultimate control over the affairs of the factory (director / partner in a company). He carries the **main legal responsibility**.
Manufacturing process	Making, altering, repairing, packing, washing, cleaning, treating or adapting any article, generating power, etc. [2(k)].
Hazardous process	Process listed in the First Schedule (e.g. chemicals, petroleum, metal extraction, fertilisers) where raw material/product may cause impairment of health or pollution.
Adult / Adolescent / Child	Adult >= 18 yrs; Adolescent 15-18 yrs; Child < 15 yrs (employment of children below 14 is prohibited).

4.2 Structure of the Act - chapter-wise scheme

Chapter (Sections)	Subject	Key provisions
II (8-10)	Inspecting staff	Inspectors appointed by the State; powers to enter, inspect, take samples, seize records, order measures.
III (11-20)	**Health**	Cleanliness, waste disposal, ventilation & temperature, dust/fume, overcrowding (14.2 m3 per worker), lighting, drinking water, latrines, spittoons.

Chapter (Sections)	Subject	Key provisions
IV (21-41)	**Safety**	Fencing of machinery (21), hoists & lifts (28), lifting machines (29), pressure plant (31), floors/stairs (32), protection of eyes (35), fire precautions (38), safety of buildings (40).
IVA (41A-41H)	Hazardous processes	Site appraisal committee, compulsory disclosure, permissible exposure limits, on-site emergency plan, workers' right to warn.
V (42-50)	**Welfare**	Washing, clothing, sitting, first-aid box, canteen, shelters, creches, welfare officers.
VI (51-66)	Working hours of adults	48 hrs/week, 9 hrs/day, rest interval, weekly holiday, overtime at double rate.
VII (67-77)	Young persons	Child < 14 not allowed; adolescents need fitness certificate; limited hours (4.5 hrs/day).
VIII (78-84)	Annual leave with wages	1 day for every 20 days worked (adult).
IX-XI	Penalties & procedure	Fines/imprisonment; general duties of occupier (7A) and manufacturers (7B).

4.3 Key safety provisions in simple words

> **Sec 21 – Fencing of machinery**: every dangerous part (moving parts, gears, belts, flywheels) must be securely fenced while in motion.

> **Sec 28 / 29**: hoists, lifts, cranes, chains, ropes and lifting tackle must be of good construction, tested and examined by a competent person (every 6 / 12 months) and marked with Safe Working Load (SWL).

> **Sec 31**: pressure plant (boilers, air receivers) must be operated within safe pressure; safety valves and examination required.

> **Sec 35**: eye protection (goggles / screens) for processes that produce flying particles or harmful radiation.

> **Sec 38**: means of escape, fire alarms, extinguishers, doors opening outwards, free passage to exits; workers trained in fire drill.

> **Sec 7A / 7B**: occupier must ensure health, safety & welfare of all workers; manufacturers/designers must test and give information on safe use of articles.

4.4 How the Act is implemented (mechanism)

1. **Registration & licensing**: the occupier gives notice, gets plans approved and a **licence** from the Chief Inspector before starting; renews it yearly.
2. **Inspection**: Inspectors visit factories, examine machines, records, registers (muster roll, accident register) and issue improvement notices / prohibitory orders.
3. **Reporting**: accidents causing death or disablement for 48 hrs or more must be reported (Form 18) - dangerous occurrences & occupational diseases too.
4. **Safety staff**: safety officers (Sec 40B), welfare officers (Sec 49), safety committees (Sec 41G) in notified factories.
5. **Enforcement**: prosecution - fine up to Rs 2 lakh and/or imprisonment up to 2 yrs for contravention; higher if death/ injury results.

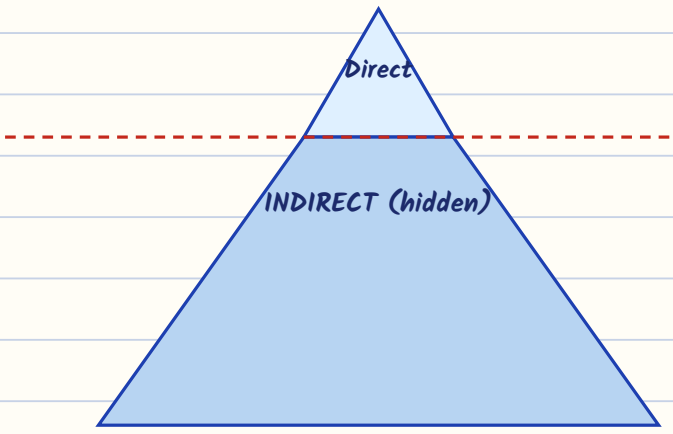
Latest update (exam tip)

> The **Factories Act, 1948** has now been subsumed by the **OSH Code, 2020** (Occupational Safety, Health & Working Conditions Code), in force from **21 Nov 2025**; the ESI Act and Employees' Compensation Act went into the Code on Social Security.

> Most safety duties (fencing, hazardous process, welfare, hours) are carried forward - JNTU/university syllabus still teaches the **Factories Act, 1948** so answer in those terms.

V. Safety and productivity

Many managers think safety **slows production**. Experience shows the opposite - **safe plants are productive plants**.



Direct (visible): medical, compensation, repair of damage - about **1 part**

Indirect (hidden): lost time, damaged plant, retraining, lower morale, legal - about **4 parts** (Heinrich)

Safety improves productivity by...	Explanation
Fewer lost hours	Less time lost in accidents, investigations, replacement of injured worker.
Better machine condition	Guarding, maintenance & housekeeping reduce breakdowns and rework.
Higher morale	Workers who feel cared for work with confidence and loyalty; lower absenteeism and labour turnover.
Lower cost	Lower insurance premium, compensation, medical bills; fewer damaged goods and penalties.
Quality & image	Orderly work-place gives better quality; good reputation attracts customers and talent.

Safety + Productivity + Quality = Profit ('**Safety is good business**')

> **Productivity** = Output / Input. Accidents reduce output and increase input (cost), so they directly reduce productivity.

> Measure of safety performance: **Frequency Rate**, **Severity Rate**, **Incidence Rate** (see formulae in Unit II).

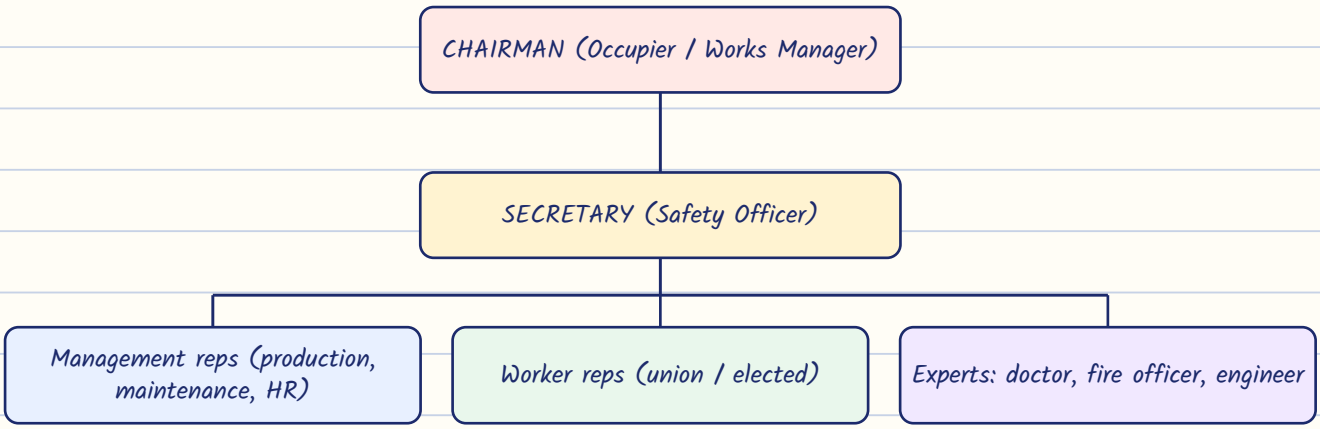
VI. Safety organizations

Organization	Level	Role / contribution
ILO	World	UN agency (1919, Geneva); sets conventions & codes (e.g. CISS OSH Convention); training, research.
OSHA / NIOSH	USA	OSHA makes & enforces standards; NIOSH does research & recommends limits.
BSC / RoSPA / HSE	UK	British Safety Council, Royal Society for Prevention of Accidents, Health & Safety Executive - training, awards, regulation.
NSC (India)	India	National Safety Council (1966, Mumbai) - promotes safety awareness, training, safety awards, National Safety Day (4 March).
DGFASLI	India	Under Ministry of Labour; advises Govt, runs Central Labour Institute (Mumbai) & Regional Labour Institutes; training, surveys.
BIS	India	Bureau of Indian Standards - IS codes for PPE, extinguishers, safety signs, machine guarding.

Organization	Level	Role / contribution
PESO	India	Petroleum & Explosives Safety Organisation - licences for explosives, petroleum, gas cylinders.
AERB	India	Atomic Energy Regulatory Board - radiation safety.
NFPA / ISO / ANSI	Intl.	Codes & standards - NFPA (fire), ISO 45001 (OHS management), ANSI (safety standards).

VII. Safety committees and structure

A **Safety Committee** is a joint body of management and workers that meets regularly to discuss safety problems and suggest remedies. Under **Sec 41G** (Factories Act) the occupier of a factory carrying on a hazardous process must set up a safety committee with **equal representation** of workers and management.



Functions of the safety committee

> Study accident reports, find causes and recommend **preventive action**.

> Conduct **safety inspections / tours** of shop floors, review hazards and near-misses.

> Suggest improvements in **work methods, PPE, housekeeping, training**.

> Organise **safety campaigns**, posters, competitions, safety week & mock drills.

> Review progress of previous recommendations, keep minutes & follow-up record.

> Act as a **two-way channel** between workers and management - build trust and participation.

Working of the committee

Point	Detail
Composition	Chairman (occupier/manager), secretary (safety officer), equal management & worker reps; doctor / fire officer as members.
Meetings	At least once a month (agenda circulated in advance); special meeting after a major accident.
Minutes	Recorded, signed, sent to departments; action points with a responsible person and a target date.
Types	Plant (central) committee, department / shop committees, contractor-safety committee.

VIII. Role of management in industrial safety

> **Top management commitment**: written **safety policy** signed by the CEO, displayed and followed - 'safety first, production next' is shown by actions, not slogans.

> **Organisation**: line managers are responsible for safety of their people; a **safety officer** (staff function) advises, trains, audits.

> **Resources**: budget for guards, PPE, training, safe layout, maintenance and emergency equipment.

> **Planning**: hazard identification & risk assessment (HIRA), safe work procedures, permit-to-work, emergency preparedness.

> **Training & motivation**: induction & refresher training, rewards for safe work, involve workers in decisions.

> **Monitoring**: inspections, audits, accident investigation, statistics, corrective action - 'plan-do-check-act'.

> **Legal compliance**: meet all statutory requirements; keep registers; cooperate with inspectors.

Line vs Staff in safety

> **Line**: supervisors & managers have authority and are **accountable** for safe work in their area.

> **Staff**: safety department **advises** (no line authority) - training, inspections, statistics, investigations.

> Under law a **Safety Officer** is required in factories with 1000 or more workers, or those with hazardous processes (Sec 40B).

IX. Role of Government in industrial safety

> **Legislation**: makes laws & rules - Factories Act, Boilers Act, Explosives Act, ESI, compensation, OSH Code.

> **Enforcement**: Factory Inspectorates, Boiler inspectorate, PESO inspect, issue notices, prosecute violators.

> **Standards & research**: BIS standards, DGFASLI / CLI studies, permissible exposure limits.

> **Training & education**: Labour Institutes, ESI hospitals, safety awards (Vishwakarma Rashtriya Puraskar), National Safety Week.

> **Social security**: compensation and ESI medical care, disability & dependant benefits.

> **Emergency preparedness**: Disaster management (NDMA), on-site/off-site emergency plans for MAH units, Public Liability Insurance Act 1991.

> **International role**: ratifies ILO conventions and coordinates with global agencies.

IX-A. Safety policy, safety officer & 3 E's

Safety policy statement

A good policy is short, written, signed by top management, displayed on notice boards and reviewed yearly. It should state: (1) the **commitment** to provide a safe & healthy workplace, (2) **responsibilities** of managers, supervisors, workers, (3) the duty to **report** hazards, (4) **compliance** with law and (5) **continual improvement**.

Sample policy lines

> 'No job is so important that it cannot be done safely.'

> 'Every employee has the right to refuse unsafe work and the duty to report it.'

> 'Safety performance will be part of every manager's appraisal.'

Duties of a safety officer (Sec 40B, Factories Rules)

> Advise management on safety laws and **safe procedures**; assist in **accident investigation**.

> Inspect the plant regularly; check guards, PPE, fire equipment, housekeeping.

> Maintain **accident records**, compute frequency & severity rates and report to the Inspector.

> Train workers (induction, tool-box talks, mock drills) and run safety weeks / campaigns.

> Act as **secretary** of the safety committee and follow up recommendations.

> Qualification: a recognised degree/diploma + a diploma in industrial safety; minimum age and experience as prescribed by the State rules.

The 3 E's of safety

E	Meaning	Examples
Engineering	Make the machine / process safe by design	Guards, interlocks, ventilation, safe layout, isolation
Education	Make the worker knowledgeable and willing	Training, posters, safety talks, demonstrations
Enforcement	Make sure rules are followed	Discipline, supervision, audits, penalties, rewards

Principles of accident prevention

1. Fix **responsibility** - line management owns safety.
2. Find **root causes** of accidents and near-misses, not just the victim's fault.
3. **Eliminate** the hazard first; use PPE only as the last line of defence (hierarchy of control).
4. Select, train and place the **right person on the right job**.
5. Keep records, analyse statistics and use them to **plan action**.
6. Involve workers - their suggestions are the cheapest hazard-detection system.

> Hierarchy of control: **Eliminate** > Substitute > Engineering > Administrative > PPE

Safety performance indices (used by all safety bodies)

Index	Formula
Frequency rate (FR)	FR = (No. of disabling injuries x 10^6) / Man-hours worked
Severity rate (SR)	SR = (Man-days lost x 10^6) / Man-hours worked
Incidence rate (IR)	IR = (No. of injuries x 1000) / Average no. of workers employed
Avg. days lost per injury	= Severity rate / Frequency rate

Eg: 12 disabling injuries, 150 days lost, 2,000,000 man-hours -> FR = 12x10^6/2x10^6 = **6**; SR = 150x10^6/2x10^6 = **75**.

Welfare & hours of work - quick table (Factories Act)

Facility / rule	Requirement
First-aid box (Sec 45)	One for every 150 workers; trained person in charge; ambulance room if 500+ workers
Canteen (Sec 46)	Compulsory if 250+ workers
Rest room / shelter (Sec 47)	If 150+ workers

Facility / rule	Requirement
Creche (Sec 48)	If 30+ women workers; for children below 6 yrs
Welfare officer (Sec 49)	If 500+ workers
Weekly hours (Sec 51)	48 hrs/week; daily 9 hrs (Sec 54); spread-over 10.5 hrs (Sec 56)
Rest interval (Sec 55)	At least 30 min after 5 hrs of continuous work
Overtime (Sec 59)	Twice the ordinary rate of wages

X. Quick revision & important questions

Remember (one-liners)

> Factory = 10+ workers with power / 20+ without power.

> Occupier carries the main responsibility under the Factories Act.

> Heinrich ratio 1:29:300; 88% accidents due to unsafe acts.

> Safety committee = management + workers (equal reps), meets monthly.

> Bhopal 1984 -> 1987 amendment (hazardous processes).

> NSC India founded 1966 - National Safety Day 4 March.

1. Trace the history and development of industrial safety in India. (10 M)
2. Explain the salient features of the Factories Act, 1948 regarding safety. (10 M)
3. Discuss the relation between safety and productivity. (5 M)
4. Describe the structure and functions of a safety committee. (10 M)
5. Explain the role of management and Government in industrial safety. (10 M)
6. Write short notes on safety organizations (ILO, NSC, DGFASLI). (5 M)

UNIT II

Accident Prevention & Protective Equipment

PPE | plant survey | body parts to protect | training | causes & cost | housekeeping | first aid | reporting | investigation | psychology

I. Personal Protective Equipment (PPE)

PPE is any equipment worn by a worker to protect against one or more hazards that cannot be removed by engineering or administrative controls. It is the **last line of defence** in the hierarchy of control - it does not remove the hazard, it only protects the wearer.

Requirements of good PPE

> Gives **adequate protection** against the specific hazard and meets the **IS / ISO / ANSI** standard.

> Light, comfortable, properly **fitting**; does not hinder movement or vision.

> Durable, easy to clean / maintain / store; reasonably cheap.

> Does not itself create a new hazard; workers are **trained** to select, use, inspect and discard it.

Types of PPE and what they protect

Body part	PPE	Hazard / remark
Head	Safety helmet (hard hat), bump cap, hair net	Falling objects, impact, electric shock (class E), entanglement of hair
Eyes & face	Safety spectacles, goggles, face shield, welding helmet	Flying chips, dust, splashes, UV/IR radiation, sparks
Ears	Ear plugs (20-30 dB), ear muffs, canal caps	Noise above 85-90 dB(A)
Respiratory	Dust mask, half/full-face respirator, SCBA, air-line respirator	Dust, fumes, vapours, toxic gas, oxygen-deficient air
Hands	Gloves - leather, rubber, PVC, cotton, chemical-resistant, heat-resistant	Cuts, burns, chemicals, vibration, electricity
Feet & legs	Safety shoes with steel toe cap, gumboots, leggings, spats	Falling/rolling objects, nails, hot metal, wet floor, chemicals
Body	Overalls, aprons, aluminised suits, acid-proof suits, hi-vis jackets	Heat, splashes, sparks, vehicles, entanglement

Body part	PPE	Hazard / remark
Fall protection	Full-body harness, lanyard, lifeline, safety net	Work at height above 2 m

Respirators – quick classification

Type	Principle	Use
Air-purifying (filter / cartridge)	Cleans the ambient air through filter or chemical cartridge	Normal oxygen (>19.5%), known contaminant at low level
Air-supplying (air-line, SCBA)	Supplies clean breathing air from outside	Oxygen-deficient, unknown or IDLH atmosphere, rescue, tanks

Selection of PPE – 5 steps

> Identify the hazard & the body part exposed (job hazard analysis).
> Check that engineering controls are not possible / not enough.
> Choose PPE with the correct **rating** (e.g. dB reduction, chemical resistance).
> Fit-test and **train** the user; provide replacements & storage.
> Inspect before each use, **supervise** wearing, replace when worn out.

II. Survey of the plant for locations (safety survey)

A **plant safety survey** is a systematic walk-through to **locate hazards** and decide where PPE or other control is needed. It is done by the safety officer with supervisors and workers' representatives.

1. Collect layout, process flow chart, list of machines, materials, chemicals and number of workers.
2. Walk through each department and note **hazard locations**: moving parts, hot surfaces, noise, dust, fumes, work at height, electrical panels, material handling routes, storage.
3. Rate hazards – probability x severity – to **prioritise** (HIRA).
4. Mark the **PPE zones** (e.g. 'Ear protection compulsory', 'Helmet area') on a plant map with safety signs.
5. Decide controls: guard, ventilate, enclose, change process; then PPE for residual risk.
6. Record findings, assign responsibility with target dates, and **re-survey** periodically.

Location in plant	Typical hazard	Protection needed
Forging / foundry	Heat, noise, molten metal	Face shield, aluminised apron, ear muffs, leather shoes
Grinding / machine shop	Flying chips, rotating parts, noise	Goggles, no loose clothes, guards, ear plugs
Welding shop	UV/IR rays, fumes, sparks	Welding helmet, gloves, ventilation, screens
Chemical / battery room	Acid splash, vapours	Acid-proof suit, goggles, respirator, eye-wash station
Store / loading dock	Heavy lifting, forklifts	Safety shoes, hi-vis vest, safe lifting training
Elevated work	Falls	Harness, scaffold tags, edge protection

Parts of the body to be protected – and the injuries

> **Head & brain** – fracture, concussion, burns -> helmet.

> **Eyes** – 'eyes are irreplaceable'; chemical burns, foreign body, blindness -> goggles / face shield.

> **Ears** – noise-induced hearing loss (permanent) -> ear plugs / muffs.

> **Lungs** – pneumoconiosis, silicosis, poisoning -> respirator, ventilation.

> **Hands & fingers** – most frequently injured (cuts, crush, amputation) -> gloves, guards, tools.

> **Feet** – crush, puncture, slips -> safety shoes.

> **Skin** – dermatitis, burns -> aprons, barrier cream.

> **Back** – strain, hernia from lifting -> training, mechanical aids, back support.

III. Education and training in safety

People cause most accidents, so people must be educated. **Training** gives knowledge + skill + attitude to work safely.

Method / stage	Content
Induction training	New workers: plant rules, hazards, emergency exits, PPE, reporting, who to call – before starting work.
On-the-job training	Supervisor teaches the safe method step-by-step (show, explain, let him try, correct).
Tool-box talks	5-10 min talk at start of shift on one topic (e.g. housekeeping, ladders).
Refresher & special	Fire fighting, first aid, confined space, crane signalling, chemical handling – periodically.

Method / stage	Content
Audio-visual & posters	Films, models, demonstrations, posters, slogans, safety notice boards, bulletins.
Campaigns	National Safety Week (4-10 March), safety contests, suggestion schemes, rewards.
Mock drills	Practice of evacuation, fire fighting, gas leak control.

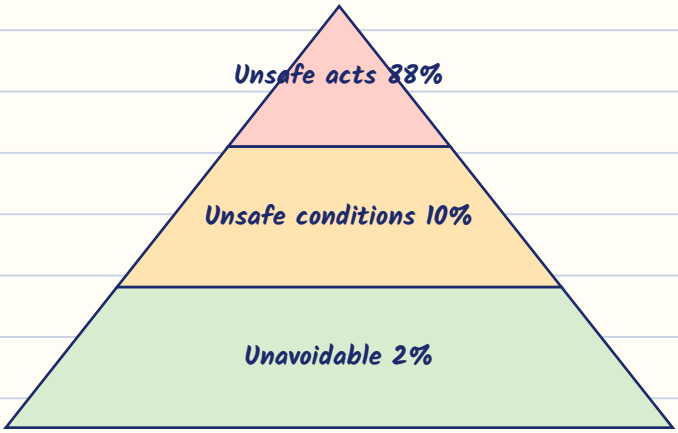
Good training: **Tell** -> **Show** -> **Do** -> **Review**

Training for supervisors is the key

> Supervisor is the **link** between management and workers - he enforces rules every day.

> Train supervisors in **job safety analysis, accident investigation, communication, leadership**.

IV. Causes of accidents



Accident causes

Direct: unsafe act or condition

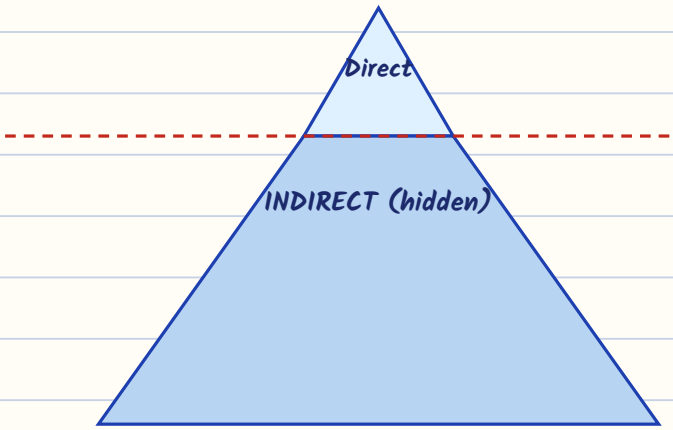
Indirect: management, planning, training, supervision gaps

Root cause analysis asks 'WHY?' five times.

Category	Examples
Unsafe acts	Operating without authority, working at unsafe speed, removing guards/safety devices, not using PPE, using defective tools, unsafe lifting, horseplay, taking shortcuts, working on moving machines, drugs/alcohol.
Unsafe conditions	Unguarded or defective machines, poor housekeeping, inadequate light / ventilation, high noise, congestion, bad layout, improper storage, missing warning signs, hazardous atmosphere.

Category	Examples
Personal factors	Lack of knowledge/skill, poor attitude, fatigue, stress, physical defects (eyesight/hearing), carelessness, alcohol.
Job / management factors	Wrong selection, poor training, inadequate supervision, no safety policy, weak maintenance, pressure on production, poor communication.
Environmental	Heat, humidity, rain, slippery floors, dust, vibration.

V. Cost of accidents



Direct (visible): medical, compensation, repair of damage - about **1 part**

Indirect (hidden): lost time, damaged plant, retraining, lower morale, legal - about **4 parts** (Heinrich)

Direct (insured) costs	Indirect (uninsured, hidden) costs
Medical & hospital expenses	Time lost by injured worker and by co-workers who stop to help / watch
Compensation & disability payments	Damage to machine, tools, material, product; repair time
Insurance premium increase	Supervisor time spent on investigation, reports, arranging replacement
Fines & legal costs	Training cost of replacement; lower output of new/injured returning worker
Funeral / dependants' relief	Loss of orders, bad publicity, lower morale, strikes, overtime

> Heinrich: **indirect cost = about 4 x direct cost**, so total cost is about 5 times the insured cost.

> Cost-benefit: Rs 1 spent on prevention saves several rupees of total accident cost.

> Human suffering of the victim and family cannot be measured in money.

VI. Housekeeping

Housekeeping means keeping the work-place clean, tidy and orderly - 'a place for everything and everything in its place'. Poor housekeeping is the cause of slips, trips, falls, fires, and obstructed escape routes.

5S	Meaning	In plant
Seiri	Sort	Remove unneeded items, scrap, old tools
Seiton	Set in order	Marked places, labels, shadow boards
Seiso	Shine / clean	Daily cleaning, oil-leak repair, waste bins
Seiketsu	Standardise	Checklists, colour codes, procedures
Shitsuke	Sustain	Discipline, audits, habit

> Keep **aisles and exits** free (yellow lines), floors dry and non-slip; clean spills at once.

> Store materials in racks, stacked stable, heavy items low; flammables in approved cabinets.

> Waste: segregated bins - oily rags in covered metal bins (spontaneous ignition), scrap removed daily.

> Good lighting, tool racks, clear signs; pipes colour-coded; machine area clean of oil/ swarf.

> Benefits: fewer accidents & fires, better productivity, space saved, good morale & image.

VII. First aid

First aid is the immediate care given to an injured or suddenly ill person **before** medical help arrives, to **preserve life, prevent worsening, promote recovery**. Law (Sec 45) requires a first-aid box per 150 workers, under a trained person.

Contents of a first-aid box (typical)

> Sterilised dressings, bandages (triangular & roller), adhesive plasters, cotton wool, gauze.

> Antiseptic lotion, burn ointment, eye wash, safety pins, scissors, tweezers, splints.

> Pain relief tablets, ORS, ammonia inhalant; first-aid manual & list of emergency numbers.

Basic first-aid actions

Situation	What to do
Bleeding	Apply direct pressure with clean pad; raise limb; do not remove soaked pad - add more; bandage; shift to hospital.
Burns	Cool with running water 10-20 min, remove rings; cover with clean non-fluffy dressing; no ice, butter, or bursting blisters.
Fracture	Do not move the limb; immobilise with splint above and below; support; treat for shock.

Situation	What to do
Electric shock	Switch OFF supply / push away with dry non-conductor; check breathing; start CPR if needed.
Chemical in eye	Wash with plenty of clean water for at least 15 min, eyelids open; cover; send to doctor.
Unconscious	Check danger, response, airway, breathing; recovery position; CPR (30 compressions : 2 breaths) if not breathing.
Poisoning / gas	Move to fresh air (rescuer with breathing apparatus), loosen clothing, oxygen, medical aid.

Golden rules of first aid

> Remain calm; make the area **safe** first (switch off power, remove from gas).

> Call for medical help early; do not give food/drink to an unconscious victim.

> Record the case in the first-aid register and report if it is an accident.

VIII. Accident reporting

Reporting gives **data** to find causes and fulfils **legal duty**. Every accident, injury, dangerous occurrence and near-miss should be reported, immediately, to the supervisor.

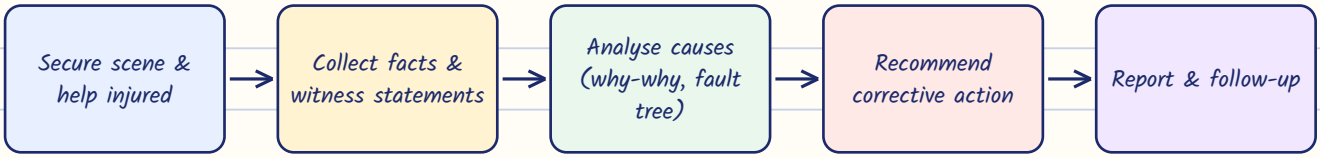
Type of report	Details
Internal report	Accident report form by supervisor within 24 hrs: who, when, where, what, injury, witness, first-aid, cause.
Notice of accident (Sec 88)	Death or injury that disables for 48 hrs or more: notify the Inspector immediately by phone/fax and send a written report (Form 18) in 12 hrs; death - within 4 hrs.
Sec 88A	Reporting of dangerous occurrences (explosion, collapse of crane, bursting of vessel, fire causing stoppage) even if no one is hurt.
Sec 89	Notice of certain occupational diseases (lead poisoning, anthrax, silicosis...) to the Inspector.
Registers	Accident book / register kept by the factory; annual returns (Form 22) to the Inspector.

> Statistics: Frequency rate, severity rate, incidence rate (see Unit I) and cost of accidents are calculated from these reports.

> A **no-blame** reporting culture increases reporting of near-misses; hiding accidents is a punishable offence.

IX. Accident investigation

Purpose: find the **root causes** to prevent repeat - **not to blame**. Start immediately while evidence is fresh.



1. **Preserve the scene**: photograph, sketch, keep failed parts, note time, weather, lighting, machine state.
2. **Interview** the injured, witnesses, supervisor separately - open questions, no pressure.
3. **Review records**: training, maintenance, permits, SOP, previous incidents.
4. **Analyse**: chain of events; 5-Why; fishbone (Ishikawa); fault tree; separate immediate / underlying / root causes.
5. **Recommend** actions with owner & date: engineering fix first, then procedure, then training.
6. **Follow-up**: verify implementation; share lessons plant-wide; update risk assessment.

Investigation team

> Supervisor of the area, safety officer, a worker representative, maintenance/engineer, and for serious cases a senior manager and external expert.

X. Industrial psychology in accident prevention

Industrial psychology studies human behaviour at work - how perception, attitude, motivation, fatigue and stress affect accidents.

Factor	Effect on accidents / remedy
Accident proneness	Some individuals have more accidents than others due to personality (impulsive, careless, risk-taker) or physical limits; remedy: correct selection & placement, counselling, job change.

Factor	Effect on accidents / remedy
Attitude & motivation	Negative attitude -> shortcuts. Motivate by participation, recognition, rewards (Maslow's hierarchy - safety is a basic need).
Fatigue	Physical or mental tiredness reduces attention, reaction; remedy: rest pauses, proper shifts, good posture, less overtime.
Monotony & boredom	Repetitive work lowers alertness; remedy: job rotation, enrichment, music, breaks.
Stress & emotional state	Worry, anger, family problems cause lapses; remedy: counselling, support, stress management.
Perception & skill	Poor vision / hearing, wrong judgement; remedy: medical checks, training, ergonomic design.
Group behaviour	Peer pressure and leadership influence safe/unsafe norms; remedy: team safety meetings and role models.

> **Selection tests**: aptitude, vision, reaction time to place right person on right job.

> **Safety attitude change**: persuasion, role-play, films, discussion; positive reinforcement works better than punishment.

XI. Safety trials & safety audits

Safety trials are planned checks to prove that a machine, process or system is safe before routine use, and to keep it safe afterwards.

Check	Purpose / how
Trial run / commissioning	New or modified machine run at no-load and load with safety devices tested, by a competent person, before acceptance.
Safety inspection / tour	Planned walk-through with checklist - daily (supervisor), weekly (department), monthly (committee), annual (management).
Safety sampling	Random observation of unsafe acts/ conditions in a sample of workers; computes % unsafe, to find trends.
Safety audit	Independent, systematic examination of the safety system against standards; gives scores and corrective-action plan.
Safety survey	Detailed study of one problem area (noise, dust, fire) using instruments.

Check	Purpose / how
Hazard & operability study	HAZOP team applies guide words (no, more, less, reverse) to process lines to find deviations.

XII. Safety related to operations

Operation	Key safety measures
Machine operation	Guards (fixed, interlocked, trip), emergency stop, no loose clothing/jewellery, switch-off & **lock-out/tag-out** before maintenance.
Material handling	Correct lifting posture (bend knees, back straight), mechanical aids, SWL marked on cranes, tested slings, no standing under load.
Work at height	Harness, scaffold inspection, guard-rails, ladders at 4:1 angle, permit for work above 2 m.
Hot work (welding/cutting)	Hot-work permit, remove flammables, fire extinguisher & fire watch, gas cylinders upright & chained.
Confined space	Entry permit, isolate, test oxygen (19.5–23.5%) & toxic gas, ventilate, standby person, rescue plan.
Electrical work	Isolate and earth, insulated tools, ELCB/RCCB, do not wet-handle, competent persons only.
Chemical handling	Labels & SDS, correct storage & segregation, PPE, eye-wash/shower, spill kit.
Pressure systems	Safety valves, periodic testing, no over-pressure, statutory inspection.

Permit-to-Work + Lock-Out/Tag-Out = control of dangerous operations

XII-A. Additional notes (important for exams)

Machine guarding – types of guards

Guard type	Working	Example
Fixed guard	Permanent barrier; tool needed to remove	Gear cover, belt enclosure
Interlocked guard	Machine cannot run unless guard is closed	Press-brake door, centrifuge lid
Adjustable guard	Adjusted to suit the job size	Band-saw blade guard, drill guard

Guard type	Working	Example
Self-adjusting guard	Moves as stock is fed	Circular saw blade guard
Trip / pull-back device	Stops the machine if hand enters danger zone	Light curtain, trip bar
Two-hand control	Both hands kept away at start of cycle	Power press

Lock-out / Tag-out (LOTO) – 6 steps

1. Inform affected persons and identify all energy sources (electric, hydraulic, pneumatic, gravity, thermal).
2. Shut down the machine in the normal way.
3. **Isolate** the energy source (open switch / valve).
4. **Lock & tag** the isolating point – each worker puts his own padlock; tag shows name, date.
5. Release stored energy (bleed, ground, block), then **verify zero energy** by trying to start.
6. After work: remove tools, refit guards, clear people, remove own lock, re-energise.

Safety colour code & signs (IS 9457)

Colour	Meaning	Example
Red	Prohibition / fire equipment / stop	No smoking, fire extinguisher
Blue	Mandatory action	Wear helmet, ear protection
Green	Safe condition, first-aid, exit	Emergency exit, first-aid post
Yellow	Warning / caution	Danger – high voltage, wet floor

Safe manual lifting

> Plan the lift, check the load and path; get help or a trolley if load > 25 kg (for men; less for women).
> Feet apart, bend knees, **back straight**, firm grip, lift with legs, keep load close to body.
> Do not twist while carrying; turn with the feet; put down in the reverse way.

Hand-tool safety

> Use the right tool for the job; hammer heads wedged, mushroomed chisels dressed.
> Spanners not extended with pipes; files with handles; screwdriver not used as chisel.
> Tools inspected, stored in racks; carried in a bag when climbing; power tools earthed and double-insulated.

Fault-tree analysis (FTA) – tool for accident analysis

FTA is a top-down, logic-diagram method: start with the **top event** (accident), then find combinations of causes joined by **AND** gates (all must occur) and **OR** gates (any one is enough) down to basic events. Probability of the top event can be calculated; the diagram shows the weakest links to strengthen.

Gate	Output occurs when...	Probability
AND	All input events occur	P = P1 x P2 x ...
OR	Any input event occurs	P = 1 - (1-P1)(1-P2)... = P1 + P2 - P1P2

Eg: fire occurs if (fuel AND oxygen AND ignition). Fuel leak P=0.1, ignition source P=0.05, oxygen is always present (1) -> P(fire) = 0.1 x 0.05 = **0.005**.

Contents of a permit-to-work

> Job description & location, date/time validity, who will do it.

> Hazards identified and precautions (isolation, gas testing, PPE, fire watch, rescue).

> Signature of issuer (authorised person), acceptor, and a final **cancel/close-out** signature.

XIII. Quick revision & important questions

Remember

> PPE = last line of defence; Hierarchy: Eliminate > Substitute > Engineering > Admin > PPE.

> Heinrich: indirect cost = 4 x direct; 88% unsafe acts.

> Accident notice: injury disabling 48 hrs / more (Sec 88), dangerous occurrence (88A).

> First aid box: 1 per 150 workers (Sec 45).

> Investigation aims at root cause, not blame.

> 5S = Sort, Set in order, Shine, Standardise, Sustain.

1. Explain the types of PPE and the parts of the body they protect. (10 M)
2. How is a plant surveyed for safety? Discuss. (5 M)
3. Discuss the causes and cost of accidents. (10 M)
4. Write about housekeeping and 5S. (5 M)
5. Describe first-aid for burns, bleeding and fractures. (5 M)
6. Explain the procedure of accident reporting and investigation. (10 M)
7. Role of industrial psychology in accident prevention. (10 M)

UNIT III

Safety Acts, Industrial Hygiene & Occupational Health

Factories Act | Explosives Act | Boilers Act | ESI Act | Workmen's Compensation | hygiene | ergonomics | diseases | noise | codes

I. Features of the Factories Act, 1948

Objective: to **regulate working conditions** in factories - **health, safety, welfare, hours of work, leave and employment of young persons**. Extends to the whole of India; applies to factories with 10+ workers (power) or 20+ (no power). Enforced by State Governments.

Feature	Provision
Health (Ch. III)	Cleanliness (Sec 11), waste disposal (12), ventilation & temperature (13), dust & fume (14), humidification (15), overcrowding (16 - 14.2 m3/worker), lighting (17), drinking water (18), latrines & urinals (19), spittoons (20).
Safety (Ch. IV)	Fencing of machinery (21), work on or near machinery in motion (22), young persons on dangerous machines (23), striking gear & belts (24), cranes & lifts (28, 29), revolving machinery (30), pressure plant (31), floors, stairs & access (32), pits & sumps (33), excessive weights (34), eye protection (35), dangerous fumes (36), explosive dust (37), fire precaution (38), safety of buildings (40).
Hazardous processes (IVA)	Site appraisal committee (41A), compulsory disclosure of information (41B), specific responsibility of occupier (41C), medical examination, permissible limits (41F), workers' participation (41G), right to warn (41H).
Welfare (Ch. V)	Washing facilities (42), storing clothing (43), sitting (44), first aid (45), canteen (46), shelters (47), creche (48), welfare officer (49).
Working hours (Ch. VI)	48 hrs/week, 9 hrs/day, spread-over 10.5 hrs, interval 30 min after 5 hrs, weekly holiday, double wages for overtime, no double employment.
Young persons (Ch. VII)	No child below 14; adolescents 15-18 need fitness certificate and token; only day shift 6 am-7 pm (children 4.5 hrs).
Women	Employment in night shift (7 pm-6 am) allowed only with consent and safeguards; no work with dangerous machines / hazardous substances in some cases.

Feature	Provision
Leave (Ch. VIII)	Annual leave with wages: 1 day per 20 days worked (adult), 1 per 15 (child).
Penalty (Ch. X)	Fine and / or imprisonment; enhanced penalty for repeated offences and where death or injury results.

Note

> Since 21 Nov 2025 these provisions are carried in the **OSH Code, 2020**. For the syllabus, Factories Act, 1948 is the reference.

II. The Explosives Act, 1884

An Act to regulate the **manufacture, possession, use, sale, transport and import** of explosives (gunpowder, dynamite, detonators, fireworks, nitrates...) in India. The present rules are **Explosives Rules, 2008**, administered by **PESO** (Petroleum & Explosives Safety Organisation), Nagpur, under DPIIT.

> **Licence** is required to manufacture, possess, sell, transport or use explosives; granted by the Chief Controller / District Magistrate.

> Explosives are classified (Class 1 divisions and safety distances): **1.1 mass explosion, 1.2 projection, 1.3 fire, 1.4 no significant hazard**.

> Safe distances between buildings (inhabited, magazine, process) are fixed in the rules; storage in **magazines** with earth-mound, lightning protection and no iron tools.

> Rules on **packing, labelling, transport** (vehicle marking, speed, escort), prohibition of smoking, fire, matches near explosives.

> Power to prohibit especially dangerous explosives; duty to **report** accidents; inspectors can enter, inspect, seize.

> Penalty: imprisonment up to 3-5 years and/or fine; offences under the Act are cognisable in some cases.

Related laws

> **Explosive Substances Act, 1908** - criminal law against misuse of explosives.

> **Petroleum Act, 1934** and **Gas Cylinder Rules** - petroleum & compressed gas storage/transport.

III. The Indian Boilers Act, 1923

Provides for the **safety of life and property** from **explosions of steam boilers** by laying down registration, inspection and certification. Rules: **Indian Boiler Regulations (IBR), 1950**. Boiler = closed vessel of capacity above **22.75 litres** that generates steam under pressure above atmospheric pressure from heat of fuel or other source.

Aspect	Provision
Authority	Central Boilers Board (makes regulations), **Chief Inspector of Boilers** in each State, Inspectors of Boilers.
Registration	A boiler cannot be used unless registered with a number & **certificate** issued after inspection; maximum working pressure is fixed.
Inspection	Annual (internal & external) inspection and hydraulic test; after repair or change; certificate valid for 12 months.
Mountings	Safety valves (2), water level gauges (2), pressure gauge, stop valve, feed check valve, blow-down valve, fusible plug, manhole.
Operation	Boiler attendants / **boiler operators** must hold a certificate of competency; not to be operated beyond rated pressure.
Accidents	Every explosion or serious damage must be reported to the Inspector; no repair of the wreckage till inspected.
Materials & repairs	Boiler materials, welding of pressure parts, welders and repairs must meet IBR; approved manufacturers.
Penalty	Use of unregistered or uncertified boiler is punishable by fine and imprisonment; explosion due to negligence - severe.

IV. Employees' State Insurance (ESI) Act, 1948

A social-security Act that gives **medical care and cash benefits** to employees in case of **sickness, maternity, employment injury, disablement and death**. Run by the **ESI Corporation** (ESIC); funded by contributions.

Point	Detail
Coverage	Factories and notified establishments with **10 or more** employees; employees earning up to **Rs 21,000** per month (Rs 25,000 for disabled).

Point	Detail
Contribution	Employer **3.25%** + employee **0.75%** of wages (total 4%); employees earning up to Rs 176/day exempt from their share.
Sickness benefit	Cash about **70% of wages** for up to 91 days in a year, plus full medical care.
Maternity benefit	Full wages for 26 weeks (first two children) and 12 weeks thereafter.
Disablement benefit	Temporary: 90% of wage during disablement. Permanent: pension based on loss of earning capacity.
Dependants' benefit	Monthly pension (about 90% of wage) to widow, children if the worker dies of employment injury.
Other	Medical benefit (for self & family), funeral expenses (Rs 15,000), unemployment allowance (Rajiv Gandhi Shramik Kalyan Yojana).

> Employee covered by ESI is **not** entitled to compensation under the Workmen's Compensation Act for the same injury.

> Administration: **ESIC** (corporation), standing committee, medical benefit council; ESI hospitals and dispensaries.

> Note: after 21 Nov 2025 the ESI Act is subsumed in the **Code on Social Security, 2020**; benefits remain broadly the same.

V. Workmen's Compensation Act, 1923

Now called the **Employees' Compensation Act (2009 amendment)**. Employer is liable to pay **compensation** if a worker suffers **personal injury by accident arising out of and in the course of employment** (resulting in death or disablement) or an **occupational disease** listed in Schedule III.

Point	Provision
Liability	Employer pays even without fault (no-fault liability) - employer not liable if the injury is due to drunkenness, wilful disobedience of safety rules or removal of safety device.
Death	50% of monthly wages x relevant factor (age-wise); minimum Rs 1,20,000.
Permanent total disablement	60% of monthly wages x relevant factor; minimum Rs 1,40,000.
Partial / temporary	Percentage of the above according to loss of earning capacity (Schedule I); temporary disablement: half-monthly payment.

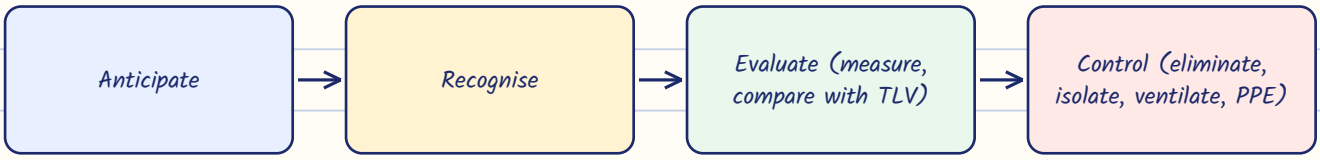
Point	Provision
Wage ceiling	For calculation, monthly wage taken is capped at Rs 8,000 (as revised by Govt).
Procedure	Claim to the **Commissioner for Workmen's Compensation** within 2 years; employer deposits compensation; penalty and interest for delay.
Medical	Employer bears medical expenses; notice of accident to be given to employer.

Compare - ESI Act vs Workmen's Compensation Act

Point	ESI Act	WC Act
Type	Social insurance (contributory)	Employer's liability (no premium)
Benefit	Medical + cash for sickness, maternity, injury	Cash on injury / disease / death
Covers	Employees up to wage ceiling	All workmen (with limits)
Administered by	ESIC	State Commissioner

VI. Industrial hygiene

Industrial hygiene is the science and art of **anticipation, recognition, evaluation and control** of environmental factors at work that may cause sickness, impaired health, discomfort or inefficiency.



Hazard class	Examples
Chemical	Dust, fumes, mists, gases, vapours - silica, lead, benzene, chlorine, CO
Physical	Noise, vibration, heat/cold, radiation, illumination, pressure
Biological	Bacteria, virus, fungi - anthrax, tetanus, TB in hides, hospitals
Ergonomic	Poor posture, repetitive motion, heavy lifting, bad workstation
Psychosocial	Stress, shift work, harassment, long hours

Threshold Limit Values (TLV, ACGIH)

> **TLV-TWA**: time-weighted average concentration for an 8-hr day / 40-hr week to which nearly all workers can be exposed repeatedly without harm.

> **TLV-STEL**: 15-min short-term exposure limit - not more than 4 times a day, at least 60 min apart.

> **TLV-C (Ceiling)**: concentration that should never be exceeded, even instantaneously.

> Units: **ppm** for gases/vapours, **mg/m3** for dust/fumes, f/cc for fibres. Factories Act 2nd Schedule lists permissible levels.

$$TWA = (C1\ T1 + C2\ T2 + \ldots + Cn\ Tn) / 8$$

Eg: 2 hrs at 100 ppm, 6 hrs at 20 ppm -> TWA = (200 + 120)/8 = **40 ppm**.

VII. Occupational safety, health and disease prevention

Occupational health (WHO/ILO): promotion and maintenance of the highest degree of **physical, mental and social well-being** of workers in all occupations.

> **Pre-employment & periodic medical examination**; special tests (lung function, audiometry, blood lead).

> **Identify and control** hazards at source; limit exposure time; job rotation.

> **Education**, personal hygiene (wash hands, no eating at work), vaccinations.

> Health records, **health surveillance**, rehabilitation and suitable job after illness.

> Occupational health centre, doctor, nurse; ambulance and tie-up with hospital.

> Levels of prevention: **primary** (remove cause), **secondary** (early detection), **tertiary** (treatment & rehabilitation).

VIII. Occupational diseases

A disease that arises mainly from work exposure - listed in **Schedule III** of the Factories Act / WC Act (e.g. 29 listed diseases).

Disease	Cause / industry	Effect & prevention
Silicosis	Free silica dust - mining, quarry, foundry, sand-blasting	Fibrosis of lungs; wet drilling, LEV, respirators, medical test
Asbestosis	Asbestos fibres - insulation, brake lining	Lung scarring, cancer; ban/replace, enclose
Byssinosis	Cotton dust - textile mills	'Monday fever', breathlessness; dust control
Coal worker's pneumoconiosis	Coal dust - coal mines	Black lung; dust suppression

Disease	Cause / industry	Effect & prevention
Lead poisoning	Lead - battery, paint, printing	Anaemia, nerve damage; substitution, hygiene, blood test
Noise-induced hearing loss	Noise > 85 dB - weaving, forging	Permanent deafness; control noise, ear plugs, audiometry
Dermatitis	Oils, acids, cement, chromium	Skin rash; gloves, barrier creams, washing
Anthrax	Hides, wool, bones - tanneries	Skin/lung infection; vaccination, disinfection
Hand-arm vibration (white finger)	Pneumatic tools, chainsaws	Numb blanching fingers; anti-vibration tools, limit time
Cadmium / benzene / mercury	Plating, chemical industry	Kidney, blood, nervous system; closed system, monitoring

IX. Ergonomics

Ergonomics (human factors) = fitting the **job to the worker** instead of making the worker fit the job - it joins anatomy, physiology and psychology with engineering design to give comfort, safety, health and efficiency.

Area	Principle / example
Workstation design	Adjustable seat, work surface at elbow height, tools within normal reach (inner 40 cm arc), foot rest.
Posture	Neutral spine, wrists straight; avoid static, twisted or overhead postures; change posture often.
Manual handling	Keep load close to body, limit weight (about 25 kg), team lift, use trolley/hoist.
Repetitive work	Rotate tasks, rest pauses, power tools with low vibration - avoid RSI, carpal tunnel syndrome.
Controls & displays	Standard colour (red stop, green start), logical layout, readable dials, proper labels, error-proofing.
Environment	Right light (200-500 lux), temperature 20-26 C, low noise and vibration.

> Benefits: fewer **musculoskeletal disorders** (MSDs), less fatigue, fewer errors and accidents, higher productivity.

> Anthropometry: design for 5th to 95th percentile of the user population.

X. Stress and fatigue

	Stress	Fatigue
Meaning	Body's response to demands/pressures that exceed coping ability	Reduced capacity to work due to prolonged mental or physical effort
Causes	Overload, deadlines, role conflict, danger, poor relations, job insecurity, shift work	Long hours, heavy work, monotony, poor sleep, heat, noise, poor nutrition
Effects	Anxiety, hypertension, ulcer, errors, absenteeism, substance use	Slow reaction, errors, accidents, sickness, low output
Control	Job design, clear roles, counselling, yoga, communication	Rest pauses, shift design, rotation, canteen food, medical check

> Physical fatigue is cured by rest; **mental fatigue** needs a change of activity / sleep; chronic fatigue needs medical attention.

XI. Health, safety and the physical environment

Factor	Effect and recommended control
Illumination	Poor light causes eye strain and accidents. General 150-300 lux; fine work 750-1000 lux (IS 3646). Avoid glare; clean luminaires; emergency lights.
Ventilation	Remove heat, fumes; natural + mechanical; air change rate 6-12/hr (workshops); local exhaust for contaminants; 14.2 m3 space/worker.
Temperature & humidity	Comfort zone 20-26 C, 40-60% RH. Heat stress: wet bulb globe temperature (WBGT) index; remedies - air showers, shielding, water & salt, rest, acclimatisation. Cold: clothing, heaters.
Radiation	Ionising (X-ray, gamma): time, distance, shielding; dosimeters; AERB limit 20 mSv/yr. Non-ionising (UV, IR, laser): screens, goggles.
Vibration	Whole-body and hand-arm; isolators, damping, job rotation.
Sanitation	Drinking water, latrines, washing, waste disposal (Sec 18, 19).

XII. Engineering methods of controlling chemical hazards

Control hazards in this order of preference - **at the source** first, then along the path, last at the worker.

Method	Description / example
Substitution	Replace hazardous material by a safer one (benzene -> toluene, asbestos -> glass wool, solvent-based -> water-based paint).
Process change	Change the process to avoid the hazard (brush painting instead of spraying, wet grinding instead of dry).
Isolation / enclosure	Enclose the process, glove boxes, remote control, distance, barriers.
Local exhaust ventilation (LEV)	Hood + duct + fan + air cleaner captures contaminant at the source; capture velocity 0.5-1 m/s.
General (dilution) ventilation	Fresh air dilutes low-toxicity contaminants; not for highly toxic or high-emission.
Wet methods	Water spray to suppress dust (wet drilling in mines).
Good housekeeping & maintenance	Prevent leaks, clean spills, vacuum (not sweep), closed containers.
Administrative	Limit exposure time, job rotation, training, labels, monitoring, medical checks.
PPE (last)	Respirators, gloves, suits when other controls are not enough.

XIII. Control of industrial noise and protection against it

Noise is unwanted sound. Measured in **decibels dB(A)** on a sound-level meter; sound level is logarithmic.

Exposure for long periods causes **noise-induced hearing loss**, stress, communication problems and accidents.

Sound level dB(A)	Permitted exposure per day (Indian rules / OSHA type, 5 dB exchange)
90	8 hrs
95	4 hrs
100	2 hrs
105	1 hr
110	30 min
115	15 min

> Action level 85 dB(A) (hearing conservation programme); Factories Act permissible limit 90 dB(A) for 8 hrs; never more than 115 dB(A).

> Typical levels: conversation 60 dB, lorry 85 dB, power press 100 dB, jet take-off 120+ dB (pain 130).

Control level	Methods
At source	Silent machine, maintenance, lubrication, balancing, vibration isolators, damping, mufflers on exhaust, replace metal-to-metal impact (rubber, plastic).
Along path	Enclosures, acoustic barriers, absorbent lining, sound-proof control rooms, increase distance, layout planning.
At receiver	Ear plugs (NRR 20-30 dB), muffs, job rotation, limit exposure time, regular **audiometry**, training.

Combining two sound levels: L = 10 log10 (10^(L1/10) + 10^(L2/10)); two 90 dB machines = 93 dB

XIV. Codes and regulations for worker safety, health and systems

Code / standard	Scope
IS 14489	Code of practice on occupational safety and health audit.
IS 14490 / IS 18001	OHS management systems (now aligned with ISO 45001).
IS 5216	Guide for safety procedures in hazardous chemical handling.
ISO 45001:2018	OH&S management system - PDCA model, leadership, worker participation, risk-based.
OSHA 29 CFR 1910	US general-industry standards - machine guarding, LOTO, hazard communication.
NFPA codes	NFPA 70 (electric), 101 (life safety), 30 (flammable liquids).
Factories Rules / OSH Code rules	State rules for safety officers, medical examination, hazardous process, licences.

Codes for safety of systems (system safety)

> **IEC 61508 / 61511**: functional safety of electrical/electronic/programmable systems; **SIL (Safety Integrity Level) 1-4** for safety instrumented systems in process industries.

> **ASME Boiler & Pressure Vessel Code, API standards**: design & inspection of boilers, pressure vessels, tanks, piping.

> **ISO 12100**: machinery safety - risk assessment & risk reduction; **ISO 13849**: control systems of machines.

> Hazard study tools: **HAZOP, FMEA, FTA, What-if, Checklist, LOPA**; Safety case & safety management for major hazard installations.

XIV-A. Worked examples & extra notes

Eg 1 - Employees' compensation (death)

A worker aged 30 earning Rs 8,000/month (wage ceiling) dies in an accident. Assume the relevant factor for his age is 200 (illustrative - take the real value from Schedule IV). Compensation = 50% x 8,000 x 200 = **Rs 8,00,000**. (Higher than the minimum Rs 1,20,000, so this amount is payable.)

Eg 2 - ESI contribution

Employee wage Rs 18,000/month: employer share 3.25% = Rs 585; employee share 0.75% = Rs 135; total = **Rs 720** per month.

Eg 3 - Noise addition

Three identical machines each give 85 dB at a point. L = 85 + 10 log10(3) = 85 + 4.8 = **89.8 dB**. So 3 machines add only about 5 dB to one.

Eg 4 - Room ventilation (air changes)

Workshop 20 m x 10 m x 5 m = 1000 m3 needs 8 air changes/hr -> airflow = 8000 m3/hr = **2.22 m3/s**; choose a fan of this capacity (add loss factor).

Types of airborne contaminants

Form	Definition	Example
Dust	Solid particles made by crushing, grinding	Silica, coal, cotton dust
Fume	Very fine solid particles from condensation of vapour (<1 micron)	Welding fume, lead fume
Mist	Liquid droplets	Acid mist, oil mist
Fog / Smoke	Condensed liquid / incomplete combustion particles	Smoke from fire
Vapour	Gaseous form of a substance that is liquid at room temperature	Benzene, acetone
Gas	Substance that is gaseous at room temperature	CO, Cl2, H2S

Routes of entry of toxic substances

> **Inhalation** (main route, ~90%), **skin absorption**, **ingestion**, injection (rare).

> Effect depends on **dose = concentration x time**, toxicity, individual sensitivity.

> Acute (short, high dose) vs chronic (long, low dose); local (irritation) vs systemic (organ damage).

Hearing conservation programme

1. Measure noise (noise survey, dosimetry) and make noise-contour map; mark areas.
2. Control noise by engineering means first.
3. Provide ear protection and train in use where >85 dB(A).
4. **Audiometric testing**: baseline and annual tests; compare for shifts.
5. Keep records and review the programme.

XV. Quick revision & important questions

Remember

> Factory: 10 / 20 workers. Overcrowding 14.2 m3 / worker. First aid 1 box / 150. Canteen 250. Creche 30 women. Welfare officer 500.

> Boiler > 22.75 L; annual inspection; IBR 1950. Explosives Rules 2008 - PESO.

> ESI: 3.25% + 0.75%; limit Rs 21,000. WC: death 50%, PTD 60% x factor.

> Noise: 90 dB(A) for 8 hrs, 5 dB exchange. TWA, STEL, Ceiling (TLV).

> Control order: substitute -> engineer (LEV) -> administrative -> PPE.

1. Explain the salient features of the Factories Act, 1948. (10 M)
2. Write short notes on Explosives Act and Boilers Act. (5 M each)
3. Discuss benefits under the ESI Act. Compare with WC Act. (10 M)
4. What is industrial hygiene? Explain TLV and TWA with an example. (10 M)
5. Explain engineering methods of controlling chemical hazards. (10 M)
6. Describe noise control and protection measures. (10 M)
7. Write short notes: ergonomics, stress & fatigue, occupational diseases. (5 M each)

UNIT IV

Fire Prevention & Protection

ignition sources | fire triangle | extinguishing principles | classes of fire | agents | extinguishers | hydrants | alarms | rescue | burns

I. Introduction – what is fire?

Fire is a rapid, self-sustaining **oxidation (combustion)** of a fuel that releases heat and light. Fire is the biggest single threat to plants – it kills people, destroys property and stops production. **Prevention** is cheaper than fire fighting.

Term	Meaning
Flash point	Lowest temperature at which a liquid gives enough vapour to flash (ignite) momentarily when a flame is brought near.
Fire point	Temperature at which vapour continues to burn after ignition (a few degrees above flash point).
Auto-ignition temperature	Temperature at which a substance ignites by itself without any external flame.
Flammable limits (LEL–UEL)	Range of vapour/air concentration that will burn: Lower & Upper Explosive Limits. Eg petrol 1.4–7.6%, LPG 1.8–9.5%, hydrogen 4–75%, acetylene 2.5–81%.
Flammable / Combustible	Flammable liquid: flash point below 65 C (IS/PESO class A < 23 C, B 23–65 C, C 65–93 C).

II. Sources of ignition

Source	Examples
Naked flames	Matches, lighters, burners, welding / cutting torch, furnaces, candles.
Electrical	Short circuit, overloaded cables, loose connections, arcing, faulty motors/heaters, sparks from switches.
Hot surfaces	Boiler, steam pipes, furnace walls, hot bearings, lamps, exhaust manifolds.

Source	Examples
Friction & mechanical sparks	Grinding, hot bearing, tools striking, belt slip, metal-to-metal contact.
Static electricity	Flow of liquids in pipes, filling tanks, conveyor belts, synthetic clothing.
Spontaneous combustion	Oily rags, coal heaps, hay, sawdust, wet grain, linseed oil.
Chemical reaction	Mixing incompatible chemicals, exothermic reactions (oxidisers + fuel), pyrophoric metals.
Lightning & radiant heat	Lightning strike, sun on lenses/glass, radiant heat from nearby fire.
Smoking & arson	Cigarette butts, deliberate fire-raising.
Compression	Rapid compression of gas (diesel engine effect, pneumatic hoses).

Fire prevention measures

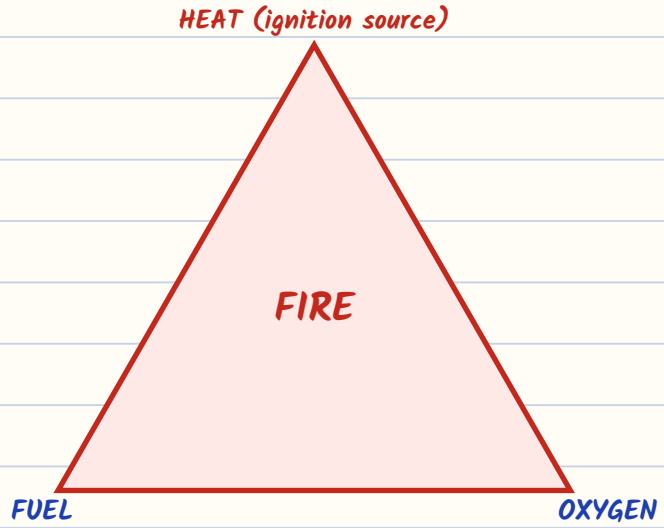
> **Control fuel**: reduce the quantity stored, keep flammables in approved containers/cabinets, away from sources; good housekeeping.

> **Control ignition**: no-smoking rules, hot-work permit, flame-proof (Ex) electrical fittings in hazardous areas, earthing & bonding to remove static, lightning conductors.

> **Control oxygen**: inerting with nitrogen in tanks; closed systems.

> Regular **inspection and maintenance** of electrics and machines; fire safety training, drills and signs.

III. The fire triangle and tetrahedron



Fire triangle = 3 elements

Remove ANY one -> fire dies

Fire tetrahedron adds a 4th:

uninhibited chain reaction

(free radicals) - this is why

halons / DCP stop fire so fast

IV. Principles of fire extinguishing

Remove one side of the fire triangle (or break the chain reaction) and the fire is put out.

Principle	How it works	Agent / method
Starvation	Remove fuel: shut valve, remove combustibles, create firebreak, drain tank	Fuel isolation, fire-break
Smothering	Exclude / dilute oxygen (below ~15%)	Foam blanket, CO2, inert gas, sand, wet blanket
Cooling	Remove heat below ignition temperature	Water jets / sprays, wet chemicals
Chain inhibition	Interrupt free-radical reaction	DCP, halons & halocarbon agents

V. Active and passive fire protection

	Active Fire Protection (AFP)	Passive Fire Protection (PFP)
Meaning	Systems that need action / motion or power to detect and fight fire	Built-in features that contain fire & maintain structure without any action
Examples	Detectors, alarms, sprinklers, hydrants, extinguishers, deluge, gas/foam systems, smoke extractors	Fire-rated walls, floors, doors, fire dampers, fire stops, intumescent coating, fire-proofing of steel, compartmentation, escape routes
Role	Detect, warn, suppress / extinguish	Delay spread, protect structure, give time for escape
Dependence	Needs power, water, human action; can fail	Always ready; no maintenance but periodic inspection

VI. Classes of fire

Class	Fuel involved	Examples	Suitable agent
A	Ordinary combustible solids (leave embers)	Wood, paper, cloth, rubber, plastics	**Water**, foam, ABC DCP
B	Flammable liquids and liquefiable solids	Petrol, oil, paint, grease, solvents, wax	**Foam, DCP, CO2**, halocarbons (not water jet)
C	Flammable gases	LPG, methane, acetylene, hydrogen	Stop gas first; **DCP**, water spray to cool

Class	Fuel involved	Examples	Suitable agent
D	Combustible metals	Magnesium, sodium, potassium, aluminium powder, titanium	**Special dry powder** (graphite, NaCl); never water
E	Electrical equipment (live)	Panels, transformers, motors, wiring	**CO2, DCP**, halocarbon; switch off first; never water/foam

Note

> Some codes use Class **F / K** for cooking oils & fats (wet chemical). Under IS 15683 electrical fire is not a separate class (it is class A/B once power is off) - but your syllabus lists A, B, C, D, E.

VII. Fire extinguishing agents

1. Water

> Cheapest, most common; works by **cooling** (high latent heat) and some smothering by steam (1 litre -> 1700 litres steam).

> Forms: **jet**, **spray / fog**, **sprinkler** (class A). Additives: wetting agents, thickeners, anti-freeze.

> **Not for**: live electrical, oil fires (spreads), metals (Na, K, Mg - releases hydrogen), burning acid.

2. Foam

> Mass of bubbles made by mixing foam concentrate + water + air; floats on liquid to **smother and cool** and suppress vapour. Used on **class B** (and A).

> Types: **Protein (P)**, **Fluoro-protein (FP)**, **AFFF** (aqueous film-forming), **FFFP**, **AR-AFFF** (alcohol-resistant) for polar solvents, **High-expansion** foam for enclosed spaces.

> Expansion ratio: low (up to 20:1), medium (20-200:1), high (200-1000:1). Mix: 3% or 6% induction.

> **Not for** live electrical, gas fires or 3-D pressure fires.

3. Dry chemical powder (DCP)

> Fine powder expelled by gas; **inhibits chain reaction** and smothers. Very fast knock-down of flames.

> **Sodium bicarbonate** (BC), **Potassium bicarbonate / Purple-K** (BC, high efficiency), **Monoammonium phosphate** (ABC, multi-purpose), Monnex (urea + K bicarbonate).

> Non-conductive; ok for electrical. Cons: no cooling (re-ignition), messy, corrosive; ABC powder damages electronics & is poor in confined space.

4. Carbon dioxide (CO2)

> Colourless, odourless, non-conductive gas stored as liquid at 50-60 bar; expands to 450 times volume; **smothers** (reduces oxygen) and cools (snow).

> Used for **class B and E** - electrical rooms, computer rooms, engines, flammable liquids. No residue.

> **Hazard**: asphyxiation in closed places above 9-10% (alarms, pre-discharge delay); not for class A deep-seated or metals (Mg burns in CO2).

5. Halons, halon alternatives, inert gases

> **Halons** (Halon 1211, 1301, 2402): hydrocarbons with bromine; best chain-breakers, clean, safe on electronics - but **ozone-depleting**, banned by **Montreal Protocol (1987)**; production stopped in 1994 (developed countries), phase-out in India by 2010.

Alternative	Examples	Features
Halocarbon (HFC)	HFC-227ea (FM-200), HFC-125 (FE-25), HFC-23, HFC-236fa	Chain inhibition + cooling; zero ozone depletion but some global-warming potential; 6-10% design conc.; electrical rooms, data centres.
Fluoroketone	FK-5-1-12 (Novec 1230)	Liquid at room temp; GWP = 1; low toxicity; 4-6% design concentration.
Inert gases	IG-541 (Inergen: N2 52%, Ar 40%, CO2 8%), IG-55 (N2+Ar 50:50), IG-100 (N2), IG-01 (Ar)	Reduce oxygen to ~12-14%; natural, no residue, no ozone/GWP; needs many large cylinders and pressure venting.
Water mist	High-pressure micro-droplets	Cooling + oxygen displacement; minimal water damage.

6. Dry powders for metal fires (class D)

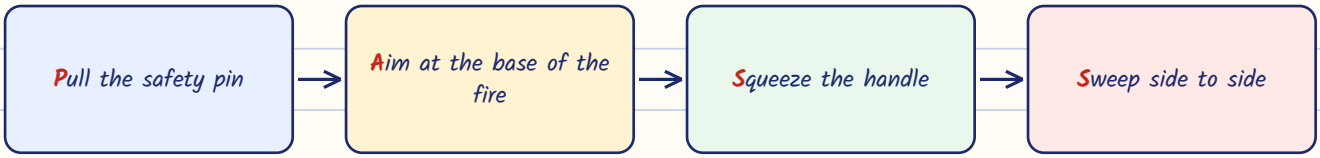
> Graphite, sodium chloride (Met-L-X), copper powder, soda ash, dry sand/dolomite - smother and form crust. Apply gently and cover; do not use water, CO2 or halogenated agents.

VIII. Types of fire extinguishers

A **portable fire extinguisher** is a cylinder containing an agent and expelling propellant; operated by hand to put out small fires. A fire must be attacked within the first **3-4 minutes**.

Type	Agent / colour band (EN 3)	Use on	Not for
Water (gas cartridge / stored pressure)	Water; red	A	B, C, D, E
Foam (AFFF)	Foam; cream	A, B	E, C, D
DCP (ABC / BC)	Powder; blue	A,B,C,E	D; confined space
CO2	Carbon dioxide; black	B, E (small A)	D, deep-seated A
Wet chemical	Potassium acetate; yellow	F (cooking oil), A	E
Clean agent (HFC)	Halocarbon; green	B, E, A	D

Operation of an extinguisher – PASS



- Stand 2-3 m upwind with escape route behind you; fires should be **small**; raise alarm first.
- Maintenance: monthly check (seal, gauge, nozzle), yearly service, hydro-test every 5 years, refill after use; mounted at 1 m height near exit, 1 per 200-250 m2 (travel distance 15-30 m).
- Fixed systems: **automatic sprinkler**, deluge, foam, CO2 total-flooding.

Fire stoppers (fire stops / firestopping)

Fire stops are **passive barriers** that seal gaps and openings around pipes, cables, ducts and joints in fire-rated walls and floors so that **flame, smoke and hot gases** cannot spread from one compartment to another.

- Materials: intumescent sealants and collars (swell on heat), mineral-wool + sealant, fire-resistant mortar, fire pillows, cable-transit blocks.
- **Fire dampers** in ventilation ducts close automatically (fusible link) when temperature reaches about 70 C.
- **Fire doors** (self-closing, rated 1-2 hr), fire-rated shutters, cavity barriers in roof/ceiling voids.

IX. Hydrant system, hoses, monitors and stand pipes

Hydrant pipes

> Underground / above-ground **ring main** (150-300 mm dia, steel/ CI) around the plant with sectional valves; fed by fire pumps (electric + diesel standby) and storage tank (static water tank).

> **Hydrants** (yard or landing valve type): pillar hydrants spaced about 30-45 m, with 63 mm outlets; each with hose box (2 hoses of 15 m, branch pipe, key).

> Pressure at the farthest hydrant: about **3.5 kg/cm2 (3.5 bar)**, flow of 2850 lpm for a main pump.

> Pipes painted **red**; tested and flushed periodically; valves exercised.

Hoses

Type	Details
Delivery hose	63 mm (2.5 in) - 15 m lengths, instantaneous couplings; **rubber-lined (RRL), canvas, PVC / polyurethane lay-flat**; type A,B,C by IS 636.
Hose reel	20-25 mm semi-rigid, 36 m, small fires, first-aid fire fighting by trained persons.
Suction hose	Hard rubber-lined, steel-wire reinforced to draw water from tank/pond.
Care	Dry after use, no kinks, roll properly, test at 15 kg/cm2 yearly, store in hose box.

Monitors

> **Fire monitor (water cannon)**: fixed or trailer-mounted nozzle with large throw (30-60 m), 1000-4000 lpm; manual / remote / oscillating; water or **foam**. Used on tank farms, refineries, loading bays.

> Placed at a safe distance so that crew need not go close to heat.

Fire watchers

> A **fire watcher** is a trained person assigned to watch for fire during **hot work** (welding, cutting, grinding) or high-risk periods.

> Duties: remove combustibles within 10 m, keep extinguisher ready, watch sparks and adjacent areas, raise alarm, and watch for **at least 30 minutes after** work ends (smouldering).

Layout of stand pipes (risers)

A **standpipe** is a vertical water pipe in a multi-storey building with outlets on every floor for fire fighting.

Type	Details
Dry riser	Empty pipe, charged by fire brigade pumps through inlet at ground floor; for buildings 15-24 m tall; 100 mm (up to 30 m) or 150 mm.
Wet riser	Permanently full of water, connected to pumps and tank; **for tall buildings > 24 m**; each floor has landing valve (63 mm) in the staircase lobby.

Type	Details
Down-comer	Wet pipe running down from the terrace tank; supplies hose on each floor by gravity; first-aid purpose.
Layout	Risers in staircase enclosures/ lobbies, one per 1000 m2 of floor area; hose reel, landing valve, 2 hoses and branch pipe; test point at terrace; combined yard hydrant + fire brigade connection.

X. Fire station, alarms and sirens

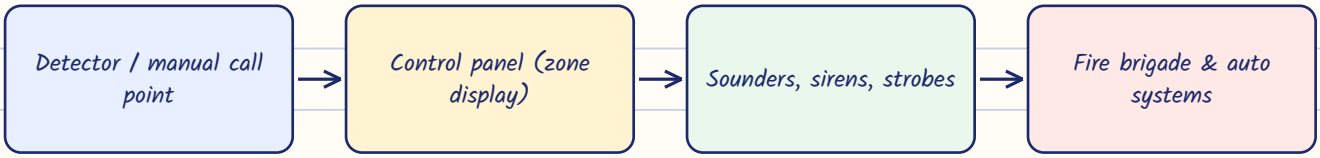
Fire station (plant fire station)

> **Location**: near the main gate/ road, central to high-risk areas, with clear access; response within 3-5 min to any point.

> Facilities: fire tender bay, water tender, foam tender, rescue van, control room (24 hrs), call receiving system, hose-drying tower, workshop, drill yard, living quarters for crew, breathing-apparatus room.

> Crew: fire officer, leaders, drivers, firemen on shift; link to the **city fire brigade** (mutual aid).

Fire alarm systems



Detector	Principle	Best for
Heat (fixed-temp / rate-of-rise)	Senses temperature (57-90 C) or fast rise; fusible element, bimetal	Kitchens, boiler rooms, dusty areas
Smoke - ionisation	Smoke reduces ion current	Fast flaming fires
Smoke - optical	Smoke scatters / obscures light beam	Smouldering fires, offices
Flame (UV / IR)	Detects radiation from flame in ms	Fuel handling, tank farms
Gas / VAD	Detects combustible gas or air sample	Process plants, data rooms

> **Manual call points** (break-glass) near exits, every 30 m; **sirens/hooters** audible 85 dB (or 65 dB above ambient) with distinct tone; **public address** and strobe lights for deaf.

> Zones, addressable systems, battery back-up, monthly test.

XI. Maintenance of fire trucks (fire tenders)

A fire tender must be **ready to move in less than 1 minute**. Preventive maintenance and daily checks are therefore compulsory.

Frequency	Checks
Daily / shift	Fuel, engine oil, coolant, battery, tyre pressure, brakes, lights & siren; water tank full; foam tank level; hoses, branch pipes, ladders, extinguishers, breathing apparatus in place.
Weekly	Start & run pump (priming, pressure, relief valve), PTO (power take-off) engagement, vacuum test, hose-reel, monitor operation, grease points.
Monthly	Test foam inductor & proportioner, check hose condition, valves leak-free, battery charge, cleaning of strainers.
Yearly	Pump performance & pressure test, hydro-test hoses and ladders, service by workshop, tyre rotation, painting & reflective marking.

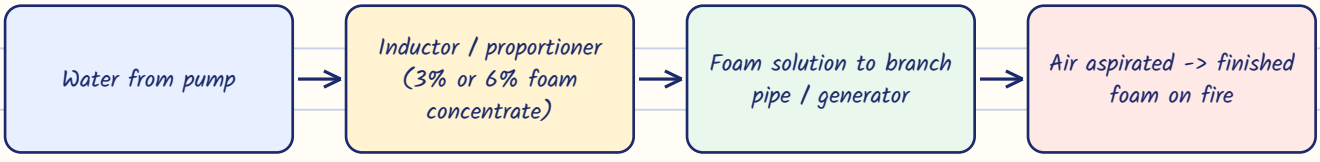
> After every use: wash and dry hoses, refill water/ foam / fuel, replace used equipment, record in the log-book.

> Records: log-book of fuel, running hours, defects, test results; trained drivers (driver-pump operators) with regular drills.

> Fire engine types: **water tender, foam tender, dry-powder tender, rescue tender, turn-table ladder, hydraulic platform, ambulance**.

XII. Foam generators

A **foam generator** (foam maker) produces foam from **foam concentrate + water + air**; the foam is then thrown on the burning liquid.



Type	Working	Use
Foam branch pipe (self-aspirating)	Water + concentrate through branch; air drawn in by venturi effect	Low-expansion foam for spill fires

Type	Working	Use
Foam generator / pourer on tank	Foam is delivered to the tank wall or by a chamber (fixed on tank top)	Fixed protection of storage tanks
High-expansion generator	A fan blows air through a screen sprayed with solution; expansion up to 1000:1	Warehouses, basements, cable tunnels, ship holds
Balanced pressure proportioner	Pump supplies concentrate in exact proportion to water flow	Refineries, fixed systems

> Foam application rates: about **4-6 L/min/m2** on hydrocarbon spill; **subsurface injection** for large tank fires; check expiry of concentrate (shelf life 8-10 yrs).

> Foam should be applied gently on the surface, from upwind, not directed into liquid.

XIII. Escape from fire & rescue operations

Escape from fire - golden rules

1. **Raise the alarm** - shout 'Fire!', operate call point, inform fire station / call 101.
2. **Leave immediately**; do not stop for belongings; help the disabled.
3. **Feel closed doors** with the back of the hand; if hot, do not open; use another route.
4. Close doors behind you to slow the fire and smoke.
5. Smoke rises: **keep low**, crawl, cover nose & mouth with wet cloth.
6. Use **stairs, never lifts**; follow green exit signs to the assembly point and report to the warden.
7. If trapped: go to room with window, seal gaps with wet cloth, signal for help. **Stop, Drop and Roll** if clothes catch fire.
8. Never re-enter a burning building.

Rescue operations

> Rescue crew wear **breathing apparatus (BA), fire-proximity suits, helmets, torch, life-line**; work in pairs, with a control point and entry-board.

> Priority order: **life -> property -> environment**. Search systematically (right-hand wall rule), mark searched rooms.

> Carry methods: drag, fireman's lift, two-hand seat, chair carry; use of ladders, ropes, escape chutes, jumping sheets, hydraulic platforms.

> Give first aid and oxygen at once and hand over to ambulance; keep a casualty record and do a head-count.

XIV. Fire drills

Fire drill = rehearsal of the emergency plan so that everyone knows what to do and how to leave in a short time. Done at least **twice a year** (once unannounced) and after any change of layout.

Element	Detail
Fire warden / marshal	Responsible persons for each area & floor; check rooms, guide staff, close doors, report to chief warden.
Assembly point	Safe open place away from the building; head-count by roll-call.
Evacuation	Alarm, shut down plant & isolate energy (if safe), leave by marked routes, disabled assisted.
Fire fighting	Fire team with extinguishers / hose reels; fire brigade informed and guided.
Review	Time taken recorded (target: building clear in 3 min), problems noted, corrections made, training updated.

Emergency organisation

> **Chief controller / Site main controller** -> Incident controller -> Fire team, rescue team, first-aid team, security, communications, evacuation & head-count team.

> On-site emergency plan is mandatory for hazardous process units (Factories Act Sec 41B) and off-site plan by the district authorities.

XV. First aid for burns

Degree	Depth / appearance	Example
1st degree	Epidermis only; red, painful, dry, no blisters	Sunburn, brief flame contact
2nd degree	Dermis; blisters, wet, very painful	Hot liquid scald
3rd degree	Full thickness; white / charred, leathery, painless (nerves destroyed)	Prolonged flame, electrical burn

Treatment - 'COOL, COVER, CALL'

1. **Stop the burning**: remove from heat; **Stop-Drop-Roll** / smother flames with blanket (not nylon).
2. **Cool**: hold under clean **running cool water for 10-20 minutes** (not ice, not very cold).
3. Remove rings, watch, belt and loose clothing **before swelling**; do not pull off clothes stuck to the

burn.

4. **Cover**: loosely with a clean, **non-fluffy, non-stick dressing** / cling film / clean cloth.
5. Treat shock: lay down, raise legs, warm; give sips of water if conscious and no vomiting.
6. **Call** medical help / ambulance (108) for large, deep burns, face, hands, genital, chemical, electrical burns, children.

DON'T

> Do not apply butter, oil, toothpaste or ointment; do not use ice; do not burst blisters; do not touch the burn with bare hands.

> **Chemical burns**: flush with plenty of water for 20 min or more (remove contaminated clothes); **electrical burns**: switch OFF power first and check heart.

Burn > 10% body area (adult) or 3rd degree = hospital

XVI. Quick revision & important questions

Remember

> Fire triangle: heat + fuel + oxygen (+ chain reaction = tetrahedron).

> Classes: A solids, B liquids, C gases, D metals, E electrical.

> Extinguishing: starvation, smothering, cooling, chain inhibition.

> Halons banned (Montreal Protocol 1987); alternatives: HFC-227ea, Novec 1230, Inergen.

> PASS: Pull, Aim, Squeeze, Sweep. Wet riser for buildings > 24 m.

> Burns: cool under running water 10-20 min, cover, call.

1. Explain the fire triangle and the principles of fire extinguishing. (10 M)
2. Classify fires and name the extinguishing agent suitable for each. (10 M)
3. Discuss water, foam, DCP and CO2 as fire extinguishing agents. (10 M)
4. What are Halon alternatives? Explain halocarbons and inert gases. (5 M)
5. Describe the types of portable fire extinguishers and their operation. (10 M)
6. Explain the hydrant system, hoses, monitors and stand pipes. (10 M)
7. Write notes on fire alarm, fire station, fire drills, rescue and first aid for burns. (5 M each)

UNIT V

Building Fire Safety

fire-safe design objectives | fire load | fire resistant materials & testing | structural protection | egress & exit width | fire certificates | high-rise

I. Objectives of fire-safe building design

Fire safety is built into a building at the **design stage**: layout, materials, structure, services and escape.

Reference code in India: **National Building Code of India (NBC 2016), Part 4 – Fire and Life Safety**, with IS codes and state fire rules.

Objective	How it is achieved
Life safety	Safe, quick escape for all occupants incl. disabled; protection of fire-fighters; smoke control.
Property protection	Limit spread to the room / compartment of origin; non-combustible construction; sprinklers.
Structural stability	Building does not collapse for the rated time so that people can leave and fire fighters can work.
Business continuity	Protect critical processes, records; quick recovery; insurance requirements.
Environment protection	Control fire water run-off & toxic smoke; avoid damage to neighbours.
Legal compliance	NBC, Fire Act/Rules of the State, Fire NOC, insurance conditions.

Basic strategy



> **Fire safety triad**: compartmentation, means of escape, fire fighting access - each independent, so that if one fails others work.

> **Occupancy classification** (NBC): Residential (A), Educational (B), Institutional (C), Assembly (D), Business (E), Mercantile (F), Industrial (G), Storage (H), Hazardous (J).

> **Types of construction** (NBC): Type I (highest fire resistance, 4 hr) to Type 4 (lowest) - depends on height, area and occupancy.

II. Fire load

Fire load is the **total amount of heat energy** that can be released by complete burning of all combustible contents of a building / compartment, including finishes, structure and furniture.

$$\text{Fire load } q = (\text{Sum of } M \times H) / A \quad [MJ/m2]$$

> **M** = mass of each combustible (kg), **H** = its calorific value (MJ/kg), **A** = floor area of compartment (m2).

> Often expressed as **equivalent kg of wood per m2**; calorific value of wood is taken as about 18-19 MJ/kg (so 1 kg/m2 wood = about 18.8 MJ/m2).

> **Fire load density** (per m2 of floor area) decides the severity & duration of a fire and so the **fire resistance** needed for the structure.

Fire load category (NBC)	Typical occupancy	Remark
Low	Dwellings, offices, hospitals, schools, hotels	Mostly furniture & paper; < about 1.1 GJ/m2 (1100 MJ/m2)
Moderate	Workshops, shops, factories, libraries, storage of non-hazardous items	About 1.1 - 2.2 GJ/m2
High	Warehouses of plastics/rubber/paper, timber, chemicals, hazardous storage	Above 2.2 GJ/m2 - need highest protection

Worked example

A store of 5 m x 4 m (A = 20 m2) holds 400 kg of wood (H = 18.8 MJ/kg), 100 kg of paper (H = 17 MJ/kg) and 50 kg of plastic (H = 40 MJ/kg). Total = 400x18.8 + 100x17 + 50x40 = 7520 + 1700 + 2000 = 11,220 MJ. Fire load = 11,220 / 20 = **561 MJ/m2** (= about 30 kg/m2 of wood) -> **low category**.

Fire severity & duration

> Equivalent fire duration (hrs) is about **fire load (kg wood/m2) / 40-50 x ventilation factor**; e.g. 30 kg/m2 -> about 0.75 hr burn at standard severity.

> More ventilation -> faster, hotter fire; less ventilation -> slower, longer fire (ventilation-controlled).

> Fire growth stages: **incipient -> growth -> flashover -> fully developed -> decay**. Flashover (about 500-600 C) is the point after which survival is unlikely.

Stage	What happens	Action
Incipient	Smouldering, little heat	Detection, extinguish with extinguisher
Growth	Flames spread, smoke layer, temperature rise	Escape, sprinklers act
Flashover	All surfaces ignite suddenly	Late - structure must hold
Fully developed	Maximum heat release	Compartment walls & structure resist
Decay	Fuel consumed; cooling	Danger of backdraught, collapse

III. Fire resistant materials

Fire resistance is the ability of a building element (wall, floor, beam, door) to **retain stability, integrity and insulation** for a stated time when exposed to fire. Rated in hours (1/2, 1, 1.5, 2, 3, 4 hr). A **non-combustible** material neither burns nor adds to fire.

Material	Behaviour in fire / use
Brick & masonry	Non-combustible; 230 mm brick wall about 4 hr; bad conductor.
Reinforced concrete (RCC)	Good: concrete cover protects steel; 2-4 hr depending on size & cover. Risk: spalling of concrete at high temperature.
Structural steel	Non-combustible but loses 50% strength at about 550-600 C -> needs protection.
Gypsum plaster / boards	Contains 21% bound water; releases as steam, keeps the temperature at about 100 C; used for ceilings, partitions.
Vermiculite / perlite / mineral-wool sprays	Light insulating coatings on steel.
Intumescent paints & sealants	Swell to a thick insulating char at 200-250 C; thin, decorative.
Fire doors & glass	Steel / timber core doors 1-2 hr rated; wired, laminated or ceramic fire glass.
Timber (treated)	Combustible but large sections char slowly (0.6-0.8 mm/min); fire-retardant treatment.
Aluminium & plastics	Aluminium melts at 660 C; plastic/PVC give smoke, toxic gases - avoid in escape routes.

Classification of materials (NBC / IS)

> **Non-combustibility test** (IS 3808 / ISO 1182): furnace test at 750 C; material is non-combustible if no flaming and a temperature rise less than 50 C.

> **Surface spread of flame** (IS 12777 / BS 476 Part 7): Class 1 (best) to Class 4 - for wall & ceiling linings.

> **Fire propagation index / ignitability, smoke density, toxicity** also considered for finishes, curtains and carpets in exits.

IV. Fire testing

Fire resistance is **proved by standard test**: a full-size element is exposed to the **standard time-temperature curve** in a furnace while loaded.

ISO 834 curve: $T = 345 \log10 (8t + 1) + 20$ (T in C, t in minutes)

Time (min)	Furnace temp (C)	Time (min)	Furnace temp (C)
5	576	60	945
15	739	120	1049
30	842	240	1153

Fire resistance - 3 failure criteria

Criterion	Meaning	Failure when
Stability (R)	Load-bearing capacity	Collapse or excessive deflection
Integrity (E)	Stops passage of flame / hot gases	Cracks, openings; cotton pad ignites
Insulation (I)	Restricts heat transmission	Average rise on unexposed face > 140 C or any point > 180 C

> Test standards: **IS 3809 (Fire resistance test of structures), ISO 834, BS 476 Part 20-22, ASTM E119, UL 263.**

> A rating '**60/60/60**' = 60 min for stability / integrity / insulation. Doors tested for **FD30 / FD60** etc. Other tests: smoke density, flame spread, fire-door and sprinkler component tests.

> **Large-scale and fire engineering**: computer models (CFD, zone models) are used for unusual buildings, with real fire scenario / design fire.

V. Structural fire protection

Methods to keep the temperature of structural members below the **critical temperature** (steel about 550 C; concrete reinforcement about 400-500 C) for the rated time.

Method	Description
Concrete encasement / cover	Steel columns or beams cast in concrete; reinforcement given 25-75 mm cover.
Boards / cladding	Gypsum, calcium silicate or vermiculite boards fixed around steel; hung ceilings and fire-rated partitions.
Spray-applied materials	Cementitious or mineral-fibre sprays on steel; fast, cheap, rough finish.
Intumescent coating	Paint expands into insulating foam char; used where appearance matters; 30-120 min.
Water-filled / concrete-filled hollow columns	Heat is absorbed by water or concrete core; used in some tall buildings.
Compartmentation	Divide building into fire compartments with fire-rated walls and floors; limits area & spread (NBC: horizontal & vertical).
Fire stopping & dampers	Seal service penetrations, ducts, shafts, cable trays.
Separation distance	Space between buildings & external wall construction to prevent spread by radiation.

Typical required fire resistance (NBC, hours)

Element	Type 1 (high-rise)	Type 2	Type 3
Load-bearing walls & columns	4	3	2
Beams	3	2	2
Floors	2	2	1.5
Non-load-bearing partitions	1-2	1	1

Note

> Values are indicative; always check the NBC 2016 Part 4 tables for the occupancy, height and area when designing.

VI. Structural integrity in fire

> Structural integrity = ability of the structure to **remain standing and avoid disproportionate (progressive) collapse** during and after fire.

> **Steel**: yield strength falls to about 50% at 550-600 C; thermal expansion pushes walls; beams sag; connections fail (WTC 2001 showed the importance of insulation & connections).

> **Concrete**: spalling (explosive flaking) due to steam pressure, loss of strength above 300 C, bond failure; use polypropylene fibres in high-strength concrete; adequate cover.

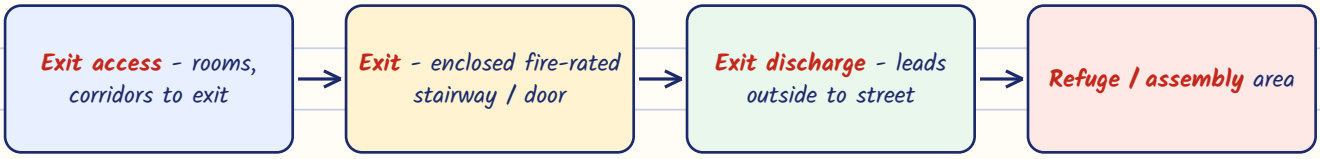
> **Redundancy and robust connections**: alternate load paths, tied structure, strong floor diaphragms, protected key elements (transfer girders, columns, stair cores).

> **Post-fire assessment**: check colour of concrete (pink above 300 C, grey above 600 C), deflection, cracks before re-occupation.

Material	Strength loss with temperature
Steel	100% at 20 C -> ~ 100% at 300 C -> ~ 50% at 550 C -> ~ 20% at 800 C
Concrete	Sound to 300 C; ~ 50% at 600 C; major loss above 800 C
Timber	Chars from outside; inner core keeps strength - predictable

VII. Concept of egress design

Egress = a continuous, unobstructed way of travel from any point in a building to a **safe place outside (public way)**. Designed so that all occupants can leave **before** conditions become untenable: **RSET < ASET** (Required Safe Egress Time less than Available Safe Egress Time).



> **RSET** = detection time + alarm / decision time + pre-movement time + travel time. **ASET** = time to reach smoke, heat, toxicity limits.

> Features: at least **two separate exits** (remote from each other), no dead-end corridors beyond about 6 m, doors open in the direction of exit, no locks that need keys.

> **Exits** enclosed in fire-resisting stair-cases (2 hr), self-closing doors, ventilation or pressurisation, direct to open air at ground.

> **Travel distance** (NBC, indicative): about 30 m for most occupancies, 45 m for industrial/storage with

sprinklers, 22.5 m for hazardous occupancy; reduce if no sprinklers.

> **Marking**: green 'EXIT' signs, floor-level photoluminescent strips, **emergency lighting** (90 min), exit signs lit by battery, plan displayed on each floor.

> **Staircase**: width, riser 190 mm max, tread 250 mm min, handrails both sides, landings; open stair and lifts are not exits.

> **Refuge areas / areas of rescue assistance** for disabled persons and for high-rise buildings; horizontal exits into the next compartment.

Egress components checklist

Component	Requirement
Doors	Min width 1.0 m, swing outward, panic bars for assembly; fire doors self-closing
Corridors	Min 1.0-1.5 m clear width (more for assembly/hospital); no obstructions
Stairs	Min 1.0-2.0 m clear width by occupancy; enclosed; 2 in number for high rise
Ramps	Slope not steeper than 1 in 12, anti-slip; for hospitals
Lifts	Not for escape in fire; except fire-lifts with special features

VIII. Exit width calculations

Exit width is calculated from the **number of occupants (occupant load)** and the **capacity of exit per unit width**. NBC uses a **unit exit width of 500 mm** (the width needed for one person stream).

Occupant load N = Floor area / Area per person ; Exit width = (N / persons per unit width) x 0.5 m

Occupancy	Floor area per person (NBC, indicative)
Residential	12.5 m2 (about)
Educational	4 m2 (classroom)
Business / offices	10 m2
Mercantile (shops)	3 m2 (ground), 6 m2 (upper)
Assembly (with fixed seats)	As per seats; unseated 0.6-1 m2
Industrial	10 m2
Storage	30 m2

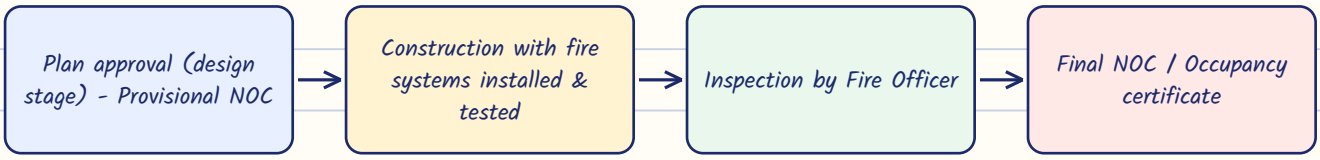
Worked example

An office floor of 600 m2. Occupant load N = 600 / 10 = **60 persons**. If **75 persons pass per unit width of door** and **50 persons per unit width of stair**: door width = (60/75) x 0.5 = **0.4 m** -> but the minimum 1.0 m is provided. Stair width = (60/50) x 0.5 = **0.6 m** -> min stair width 1.0 m governs -> provide **1.0 m, two stairs**.

> Check the **total capacity of all exits** when the largest exit is blocked (assume one exit not available).
> Width needed grows with the number of floors served: stair width at lower floors must accumulate the occupants of upper floors.
> Use of simulation (evacuation models like Pathfinder / buildingEXODUS) for large assembly and high-rise.

IX. Fire certificates (Fire NOC)

A **Fire No-Objection Certificate (NOC)** is issued by the **State Fire Service / Fire Officer** to show that the building meets the fire-safety norms. Mandatory for high-rise (15 m or more), assembly, institutional, industrial, hazardous and many other buildings; the exact thresholds are set by the State.



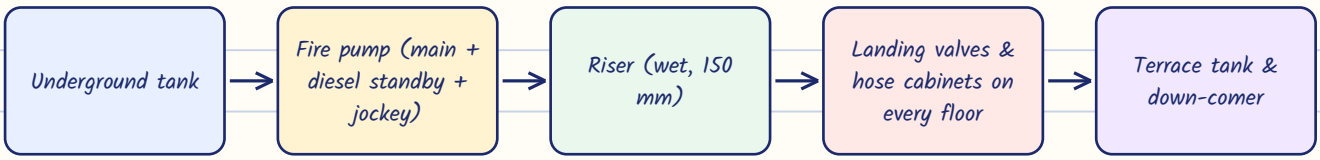
> Documents: building plans, fire-fighting layout, details of water tanks & pumps, escape routes, fire-protection design, structural-stability certificate, fire-system test reports & installer certificates, insurance.
> **Validity & renewal**: usually valid for 1-3 years (varies by State); **renewal** after re-inspection; fee; changes in use/ extension need a fresh NOC.
> **Fire insurance and licence**: occupancy/trade licence, factory licence, PESO licence often need the NOC. The Owner/Occupier is responsible for maintenance.
> Annual **fire-safety audit** by an accredited agency; mock drill logs; maintenance records.
> Non-compliance: prohibitory order, sealing, penalty, prosecution (e.g. under the State Fire Prevention and Fire Safety Acts).

X. Fire safety requirements for high-rise buildings

High-rise building: height **15 m or more** (NBC; some states use 18 m / 24 m) - fire-fighting ladders can't reach from outside, so internal systems are essential. Features follow NBC 2016 Part 4.

Requirement	Details (indicative)
Staircases	At least two (one may be external) fire-resisting enclosed stairs (2 hr); min width 1.5 m (1.0 m residential); **pressurised** (50 Pa) to keep out smoke.
Fire lift	At least 1 per 8 floors / compartment; lift lobby; capacity 545 kg (8 persons); fire-fighter's switch; speed to reach top within 1 min; standby power.
Refuge areas	Safe areas on floors for temporary shelter (about 15 m2 per 24 m of height - as per NBC) and for physically disabled.
Wet riser / down-comer	Wet riser in staircase lobbies with landing valves; 100-150 mm dia; fed by fire pumps; down-comer from terrace tank.
Fire pumps & water storage	Underground tank (about 50,000 - 2,00,000 L) + terrace tank (20,000-50,000 L); electric + diesel pump 2850 lpm; jockey pump.
Automatic sprinklers	Required in basements, high-risk areas and in buildings above 24-30 m; activated at 68 C (bulb).
Detection & alarm	Addressable smoke/heat detectors, manual call points, public address and voice evacuation, fire control room on ground floor.
Smoke management	Smoke extraction (basement 12 air changes/hr), pressurisation of stairs & lobbies, smoke curtains/shafts.
Compartmentation	Fire-rated floors, shafts, vertical ducts; fire dampers; fire stopping of services.
Emergency lighting & power	Standby DG set, exit signage, lights in stairs for 90 min.
Access for fire engines	Fire-fighting vehicle access around the building, 6 m wide with turning, 45 tonne loading; hydrants around; helipad for very tall.
Management	Fire officer, fire warden, evacuation plan, fire drills every 6 months, signage, training, annual NOC renewal.

Typical high-rise fire-fighting system layout



X-A. Supporting topics

Automatic sprinkler system

Type	Working	Use
Wet pipe	Pipes always full of water; bulb bursts at 68 C and water flows	Most buildings (simplest, most reliable)
Dry pipe	Pipes hold air under pressure; valve opens when sprinkler operates	Unheated places (freezing risk)
Pre-action	Needs detector signal + sprinkler operation	Data centres, museums
Deluge	Open sprinklers; all discharge together on detection	High hazard: aircraft hangar, transformers

> Bulb colour shows rating: orange 57 C, red 68 C, yellow 79 C, green 93 C, blue 141 C.

> Coverage about 9-12 m2 per sprinkler (light hazard); design density 2.25 - 12.5 mm/min depending on hazard class.

Smoke - the main killer

> Most fire deaths (about 60-80%) are due to **smoke and toxic gases (CO, HCN)**, not burns; smoke also blocks visibility so people lose their way.

> **Stack effect** in tall buildings drives smoke upward through shafts and stairs, so lift shafts and stairs need pressurisation or smoke lobbies.

> Smoke control: compartments, smoke curtains / screens, natural or mechanical **smoke extraction**, vents at the top, make-up air.

Fire safety management in buildings

1. Appoint a **fire safety officer** and fire wardens; prepare an emergency evacuation plan.
2. Carry out a **fire risk assessment** regularly and keep records.
3. Maintain fire systems with **inspection - testing - maintenance (ITM)** schedule: alarm weekly, extinguishers monthly, sprinklers quarterly, pumps weekly run.
4. Keep exits unlocked and clear; no storage in stair-cases; check fire doors are not wedged open.
5. Train staff and conduct fire drills; give induction to new occupants and visitors.
6. Control hot work, electrical load, waste, and decorative materials; use fire-retardant furnishings.

Lessons from major building fires

Fire	What went wrong
Triangle Shirtwaist, NY (1911)	Locked exits, blocked stairs; 146 deaths - start of fire code.
Uphaar Cinema, Delhi (1997)	Transformer fire, smoke in balcony, blocked exits and extra seats; 59 died.
AMRI Hospital, Kolkata (2011)	Basement stored flammables; smoke spread, locked windows, no alarm; 90+ died.
Kamala Mills, Mumbai (2017)	Illegal rooftop restaurants, flammable decor, no NOC; 14 died.
Grenfell Tower, London (2017)	Combustible cladding, single stair, no sprinklers; 72 died.
Takshashila Arcade, Surat (2019)	Fire at coaching centre with no exits; students jumped; 22 died.

Safe building = **Compartment** + **Escape** + **Detection** + **Suppression** + **Management**

Egress example 2 - assembly hall

A hall for 600 persons: exits for assembly at 100 persons per unit width of door (0.5 m): width = (600/100) x 0.5 = **3.0 m total**. Provide **3 exits of 1.5 m** (so that loss of one still leaves 3.0 m). Stairs at 50 persons per unit: (600/50) x 0.5 = 6 m of stairs.

XI-A. Active & passive systems in buildings - summary

Passive (built-in)	Active (operated)
Fire-rated walls, floors, shafts	Detectors, alarm, voice evacuation
Fire doors, dampers, fire stops	Sprinklers, deluge, gas / foam systems
Protected stairs, corridors, refuge	Hydrants, wet riser, hose reels, extinguishers
Structural fire protection, cladding choice	Smoke extraction, stair pressurisation, fire lift
Separation distance, fire-resistant roofs	Emergency lighting & standby generator

Fire safety signs (IS 12349 / ISO 7010)

Sign colour / shape	Meaning	Example
Green square	Escape route, exit, first aid	Running man + arrow
Red square	Fire-fighting equipment	Extinguisher, hydrant, call point

Sign colour / shape	Meaning	Example
Red circle with bar	Prohibition	No smoking, no naked flames, do not use lift in fire
Blue circle	Mandatory	Keep fire door closed

Types of construction (NBC) in short

> **Type 1**: all structural parts non-combustible, fire resistance 4 hr (columns), 3 hr (beams) - high-rise, hospitals.

> **Type 2**: 3 hr columns, 2 hr beams. **Type 3**: 2 hr columns, 2 hr beams. **Type 4**: 1 hr, mainly timber / low-rise.

> Choice depends on **height, floor area and occupancy hazard**; large buildings with higher fire load demand higher type.

Quick formulas of the unit

Formula	Use
q = SUM(M x H) / A	Fire load density (MJ/m2)
T = 345 log10 (8t + 1) + 20	ISO 834 standard fire temperature
N = Area / Area per person	Occupant load
W = (N / persons per unit) x 0.5 m	Required exit width
RSET < ASET	Safe egress condition

XI. Quick revision & important questions

Remember

> Fire load = sum(M x H) / A MJ/m2 (low / moderate / high).

> Fire resistance: stability, integrity, insulation; ISO 834 curve; steel critical about 550 C.

> Egress: exit access, exit, exit discharge; RSET < ASET; at least 2 exits.

> Unit exit width 500 mm; occupant load = area / area per person.

> High-rise = 15 m +; 2 stairs, fire lift, wet riser, sprinkler, refuge, fire NOC.

1. State the objectives of fire-safe building design. (5 M)
2. Define fire load. How is it calculated? Classify occupancies by fire load. (10 M)

3. Explain fire-resistant materials and fire resistance testing. (10 M)
4. Describe methods of structural fire protection and structural integrity. (10 M)
5. Explain the concept of egress design and calculate the exit width for a given building. (10 M)
6. What are fire certificates? Procedure to obtain Fire NOC. (5 M)
7. Discuss fire safety requirements for high-rise buildings. (10 M)